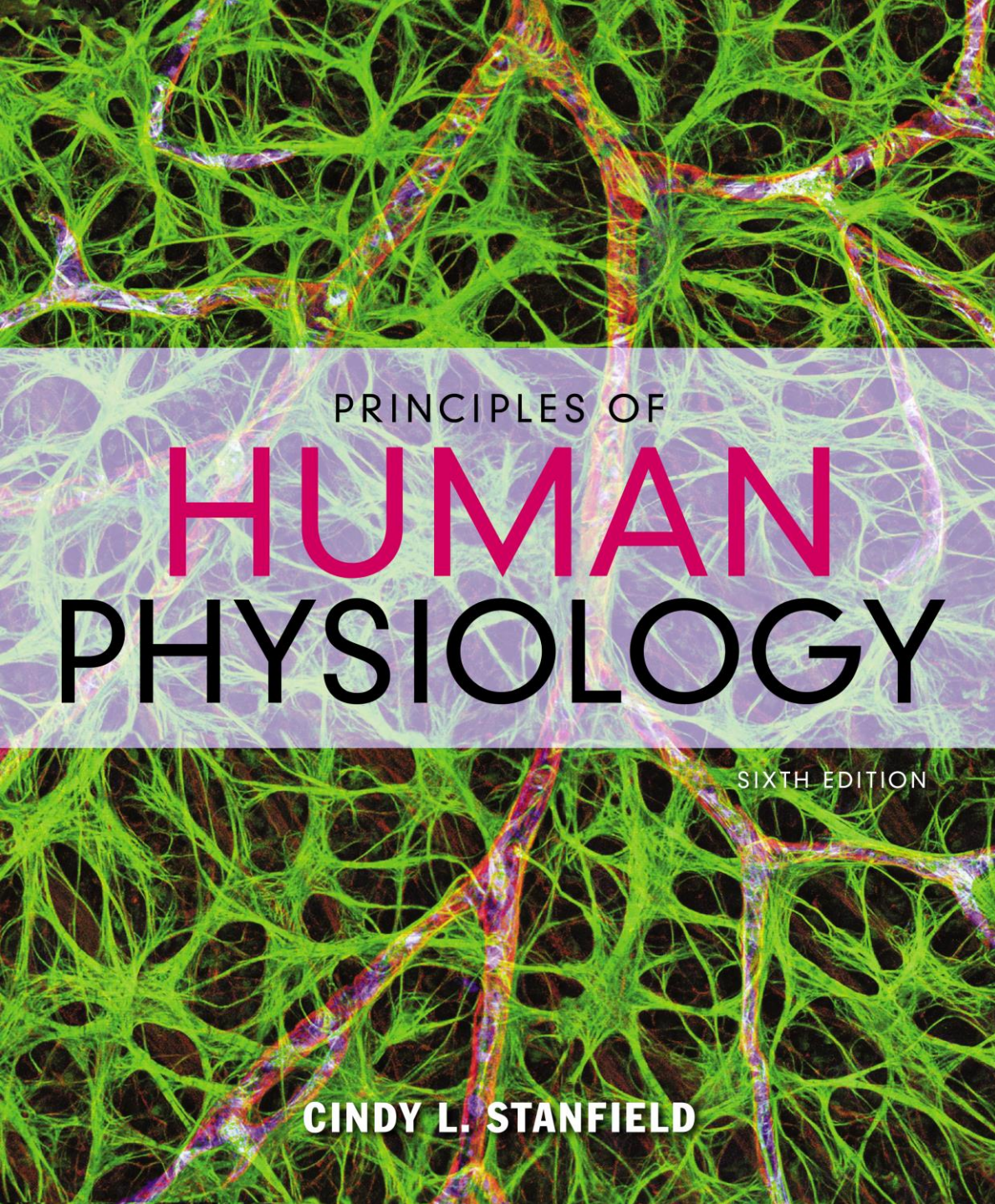




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PRINCIPLES OF
HUMAN
PHYSIOLOGY

SIXTH EDITION

CINDY L. STANFIELD

CHAPTER **20a**

The
Gastrointestinal
System

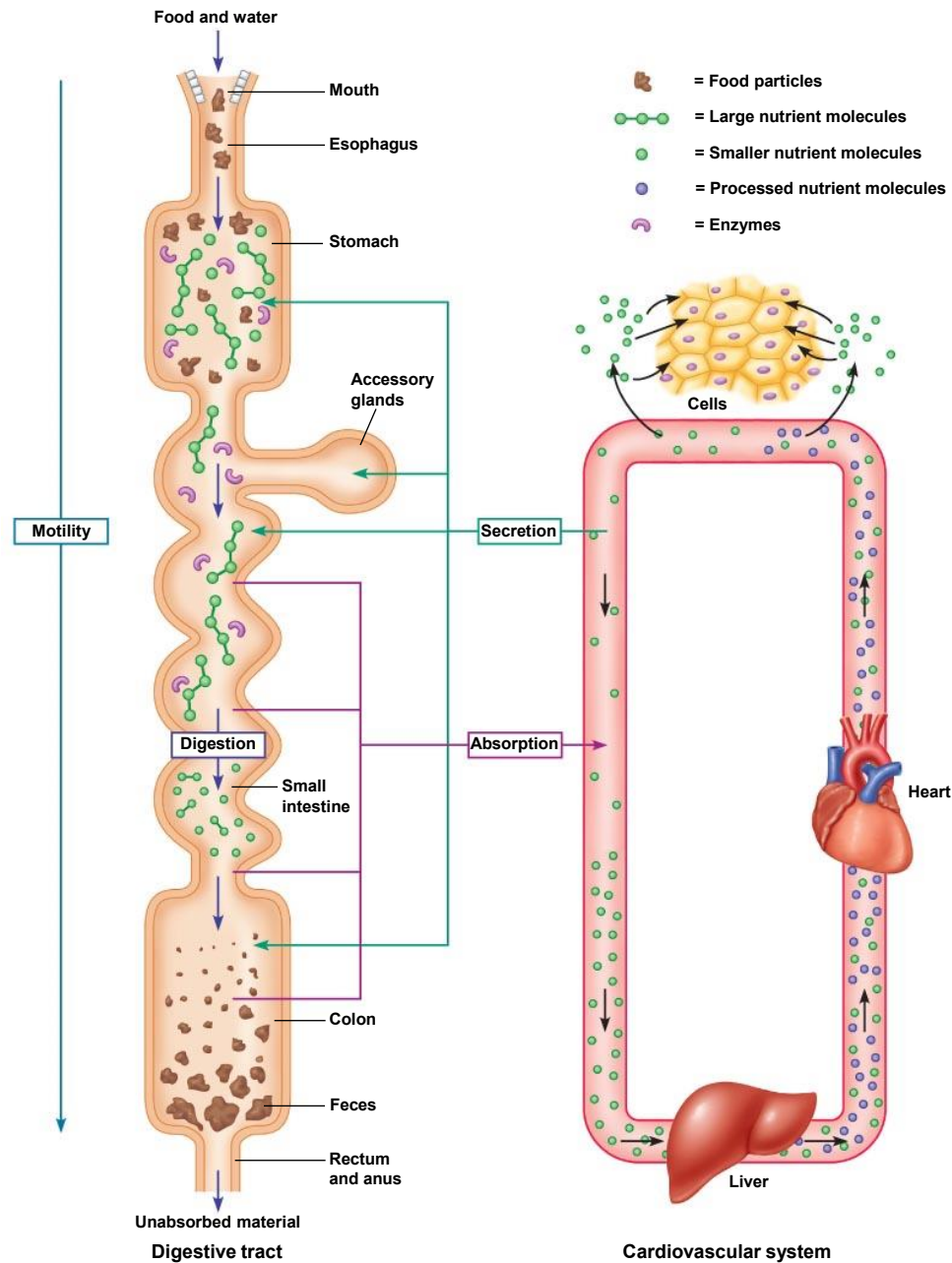
Chapter Outline

- 20.1 Overview of Gastrointestinal System Processes
- 20.2 Functional Anatomy of the Gastrointestinal System
- 20.3 Digestion and Absorption of Nutrients and Water
- 20.4 General Principles of Gastrointestinal Regulation
- 20.5 Gastrointestinal Secretion and Its Regulation
- 20.6 Gastrointestinal Motility and Its Regulation

20.1 Overview of Gastrointestinal System Processes

- Consumed nutrients are too large to enter directly into the bloodstream
- Nutrients undergo digestion before absorption
- Motility and secretion aid digestion and absorption

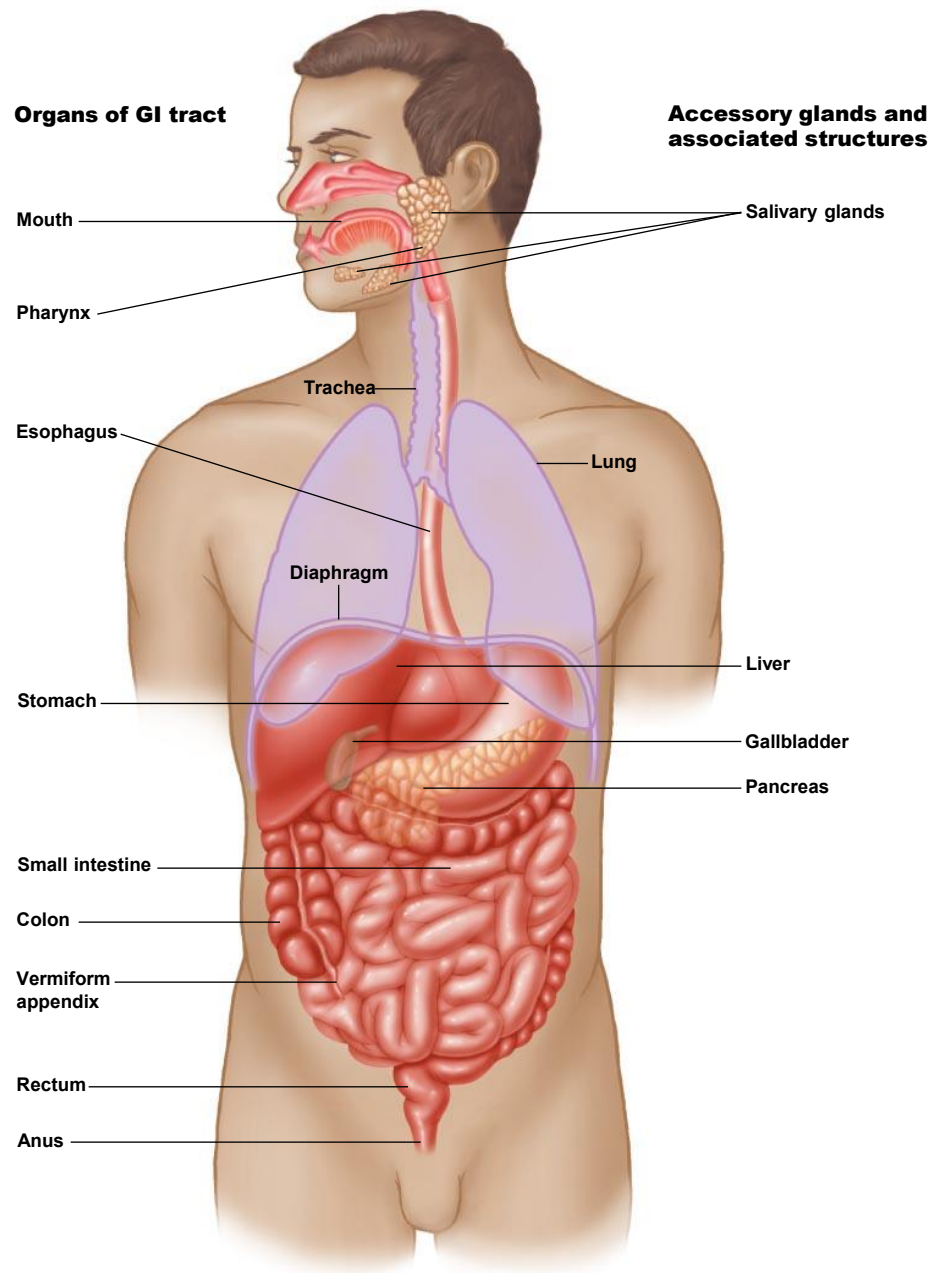
Figure 20.1 Overview of the four basic digestive processes: digestion, absorption, secretion, and motility.



20.2 Functional Anatomy of the Gastrointestinal System

- Divisions of the gastrointestinal system
 - Gastrointestinal (GI) tract
 - Hollow tube, 15 feet long
 - Mouth → pharynx → esophagus → stomach → small intestine → colon → rectum → anus
 - Accessory glands
 - Several glands outside the GI tract secrete fluids and enzymes into the lumen of the GI tract

Figure 20.2 Major structures of the gastrointestinal system.



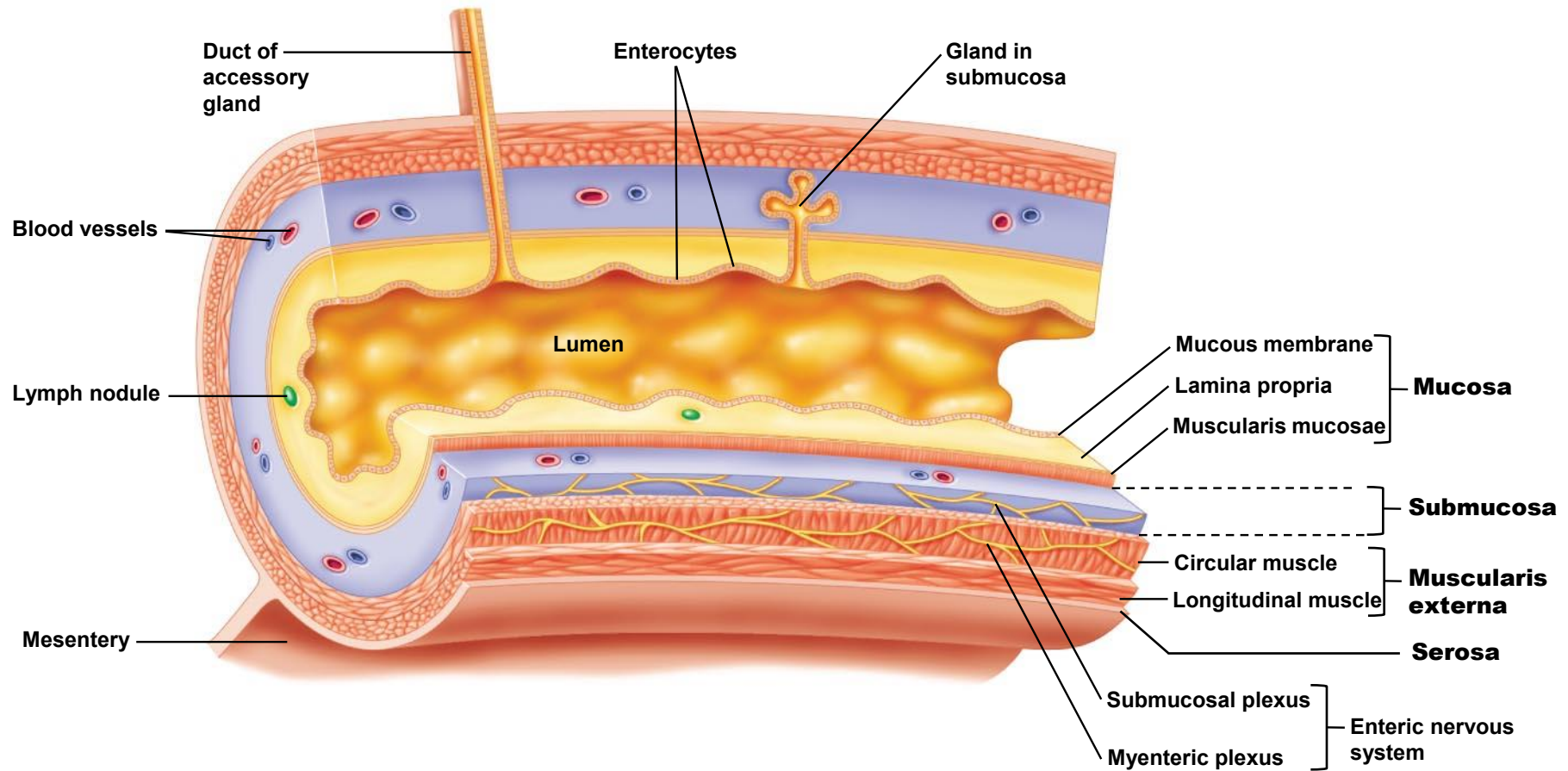
The Gastrointestinal Tract

- Wall of GI tract
 - Mucosa
 - Lining of the lumen
 - Submucosa
 - Connective tissue
 - Muscularis externa
 - Smooth muscle
 - Serosa
 - Connective tissue

The Gastrointestinal Tract

- Mucosa: mucous membrane
 - Separates GI lumen from internal environment
 - Enterocytes: epithelial cell types
 - Absorptive cells
 - Exocrine cells
 - Endocrine cells
 - Three layers
 - Epithelial lining
 - Lamina propria: connective tissue
 - Muscularis mucosae: thin smooth muscle

Figure 20.3 Generalized structure of the gastrointestinal wall, depicting the four major layers and the tissues within them.



The Gastrointestinal Tract

- Mucosa: lamina propria
 - Connective tissue
 - Small blood vessels and lymphatic vessels
 - Nerves
 - Lymphoid tissue
 - Lymph nodules
 - Peyer's patches

The Gastrointestinal Tract

- Mucosa: muscularis mucosae
 - Thin layer of smooth muscle
 - Longitudinal fibers
 - Circular fibers
 - Functions in mixing lumenal contents

The Gastrointestinal Tract

- Submucosa
 - Thick layer of connective tissue
 - Distensibility
 - Elasticity
 - Large blood vessels and lymphatic vessels
 - Submucosal plexus (Meissner's plexus)
 - Part of enteric nervous system

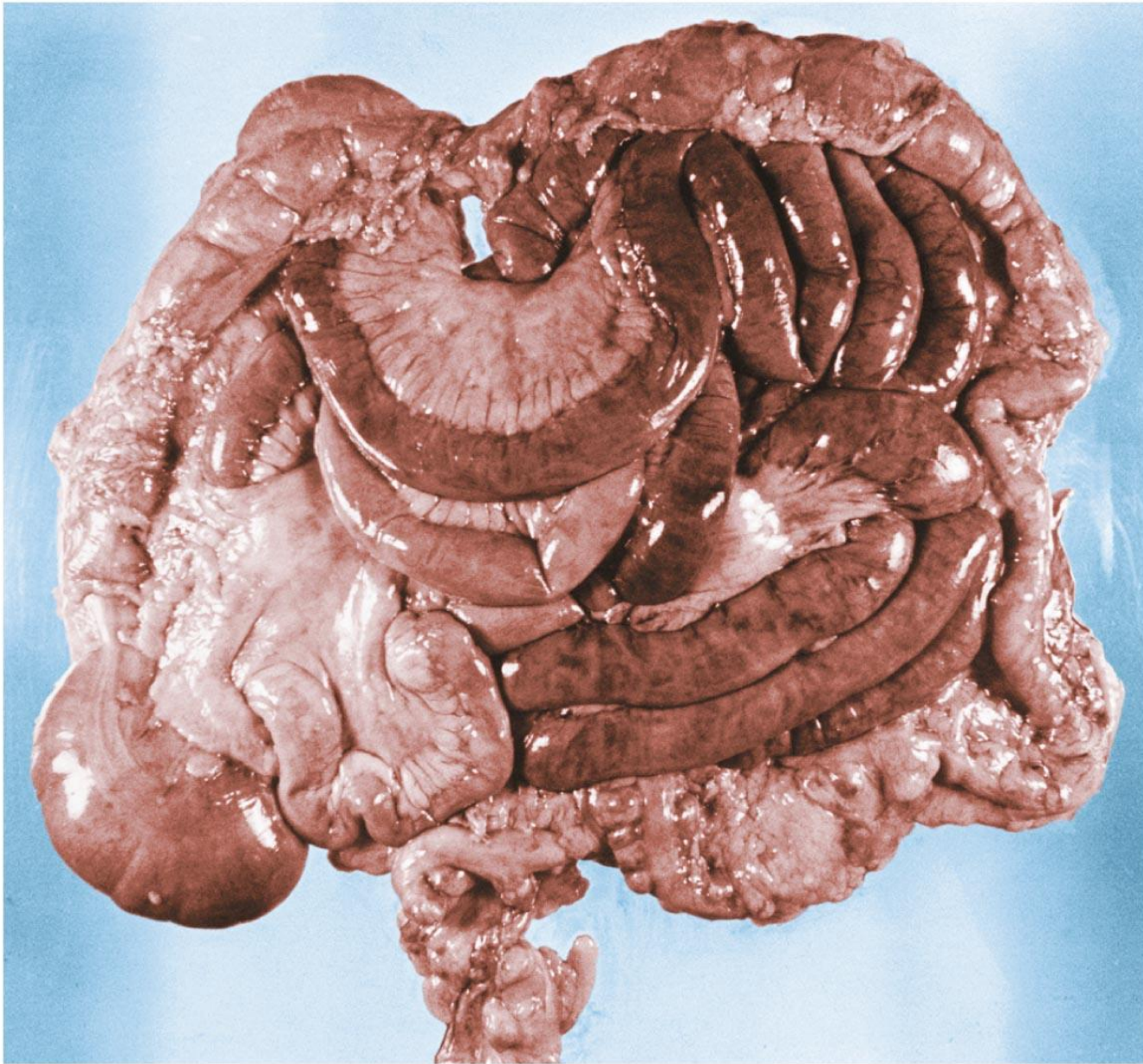
The Gastrointestinal Tract

- Muscularis externa
 - Two layers of smooth muscle
 - Inner circular layer: changes diameter
 - Outer longitudinal layer: changes length
 - Myenteric plexus (Auerbach's plexus)
 - Second part of enteric nervous system

The Gastrointestinal Tract

- Serosa
 - Inner layer: connective tissue
 - Outer layer: epithelial tissue (mesothelium)
 - Mesothelium is continuous with mesenteries, which are continuous with peritoneum

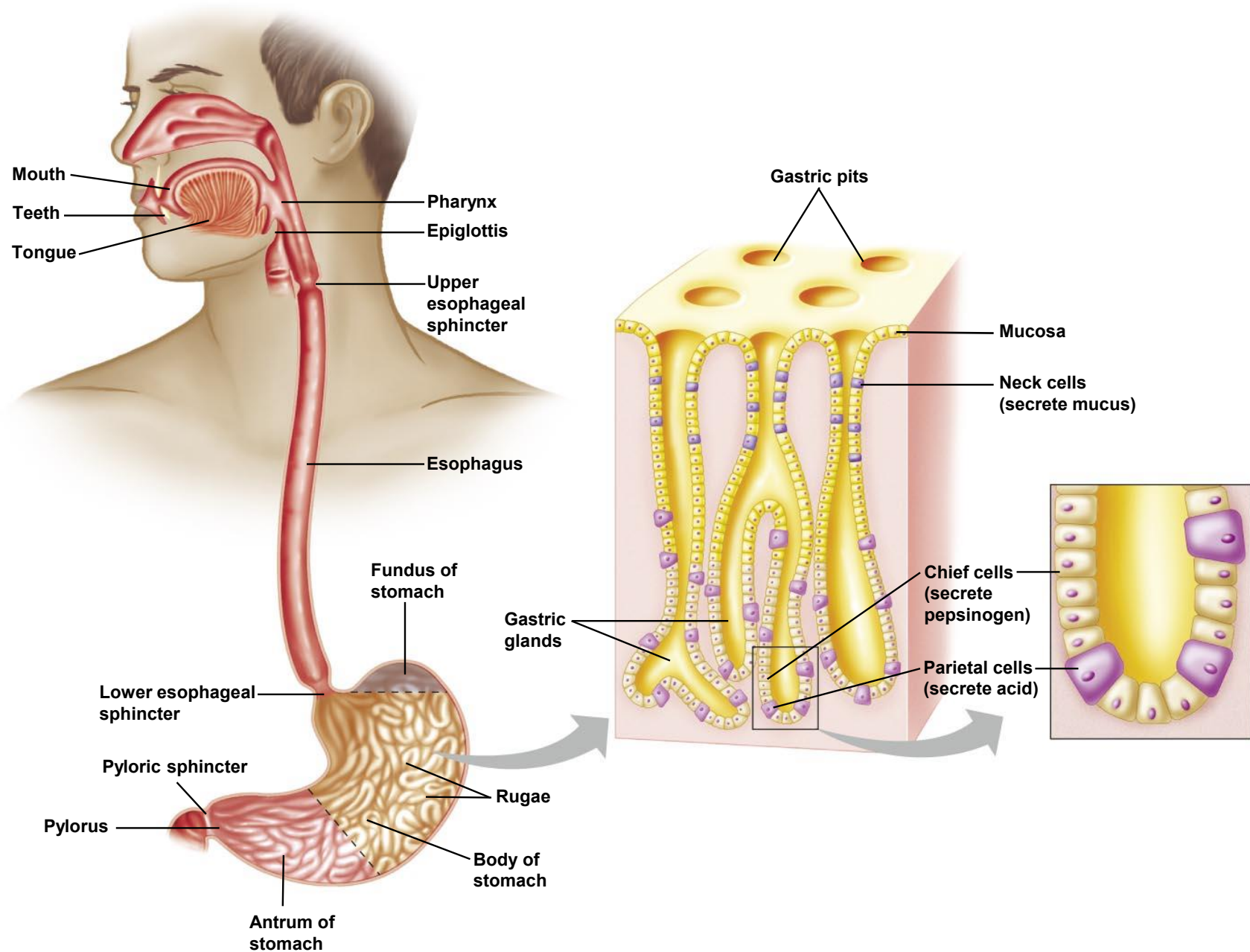
Figure 20.4 Mesenteries.



The Gastrointestinal Tract

- Functional anatomy of gastrointestinal tract organs
 - Mouth, pharynx, and esophagus
 - Stomach
 - Small intestine
 - Colon
 - Rectum and anus

Figure 20.5 Anatomy of the mouth, pharynx, esophagus, and stomach.



The Gastrointestinal Tract

- Mouth
 - Digestion begins with mastication (chewing)
 - Secretion: saliva
 - Lubricant
 - Salivary amylase: digests starch and glycogen
- Pharynx
 - Throat
 - Common passageway for air and food
 - Food enters via the esophagus

The Gastrointestinal Tract

- Esophagus
 - Muscular tube from pharynx to stomach
 - Upper one-third: skeletal muscle
 - Lower two-thirds: smooth muscle
 - Upper esophageal sphincter
 - Skeletal muscle
 - Between pharynx and esophagus
 - Lower esophageal sphincter
 - Smooth muscle
 - Between esophagus and stomach

The Gastrointestinal Tract

- Stomach
 - Stores food after it is swallowed
 - Empty volume: 50 mL
 - Full volume: 1000 mL
 - Secretes gastric juice
 - Releases food into intestine slowly

The Gastrointestinal Tract

- Stomach
 - Fundus
 - Body
 - Antrum
 - Contractions
 - Mix chyme, which is passed through pylorus
 - Gastric emptying
 - Stomach folds
 - Stomach expands to accommodate food as swallowed
 - Rugae: folds that flatten on expansion

The Gastrointestinal Tract

- Gastric pits
 - Located in stomach lining; lead to gastric glands
 - Secretory products
 - Pepsinogen: secreted by chief cells
 - Precursor for pepsin, enzyme that digests proteins
 - Hydrogen ions: secreted by parietal cells
 - Maintain acidic environment of stomach
 - Intrinsic factor: secreted by parietal cells
 - Necessary for absorption of vitamin B₁₂
 - Gastrin: secreted from G cells (hormone)
 - Mucus: secreted from neck cells

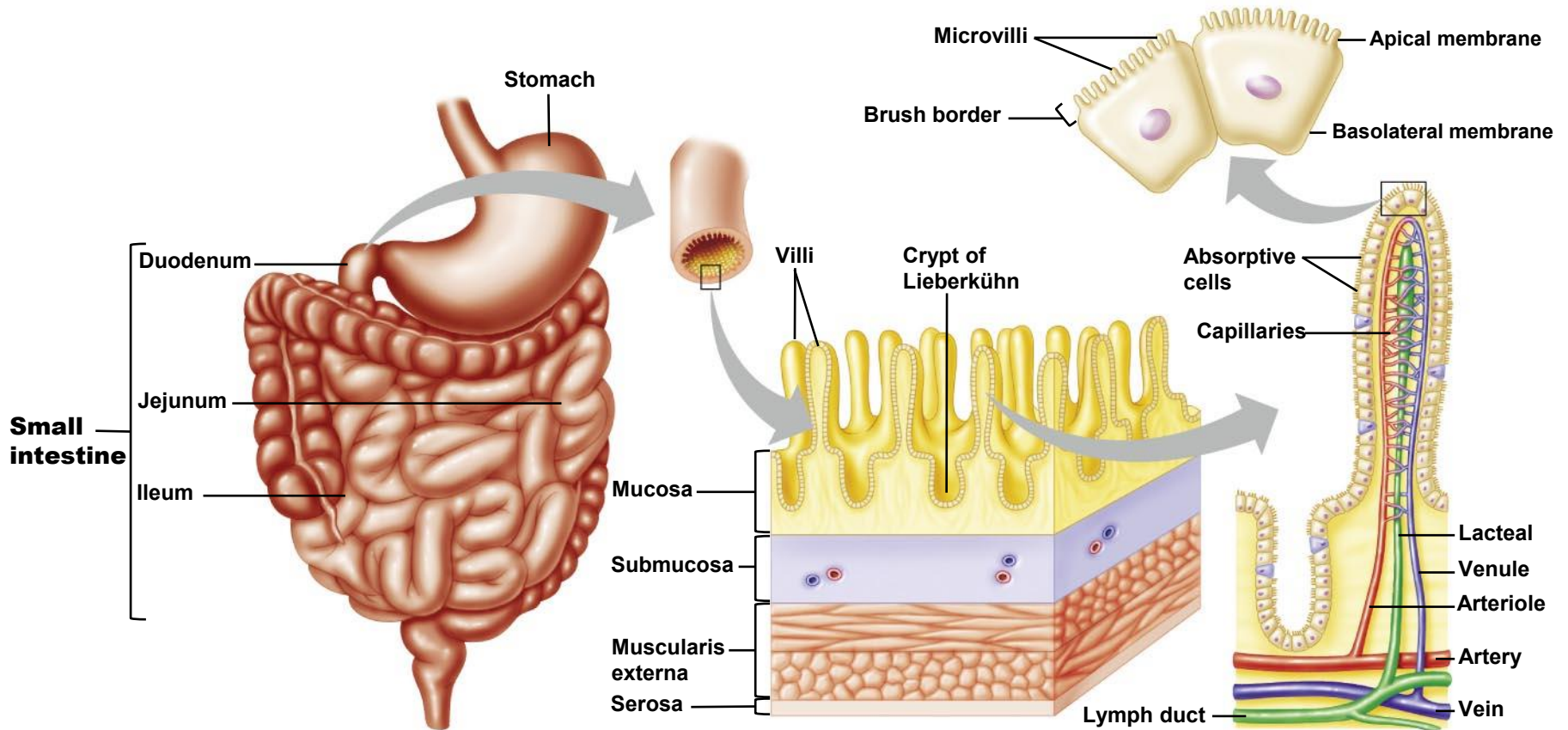
The Gastrointestinal Tract

- Acidic environment of stomach
 - pH = 2
 - Necessary for activating pepsinogen
 - Denatures proteins
 - Kills bacteria
 - Gastric mucosal barrier
 - Protective layer of mucus and bicarbonate
 - Secreted from neck cells and goblet cells

The Gastrointestinal Tract

- Small intestine
 - Coiled, hollow tube
 - 8–10 feet long
 - Between stomach and large intestine
 - Primary site of digestion and absorption
 - Three divisions
 - Duodenum
 - Jejunum
 - Ileum

Figure 20.6 Anatomy of the small intestine.



The Gastrointestinal Tract

- Secretions into duodenum
 - Pancreatic juice enters duodenum
 - Digestive enzymes
 - Bicarbonate: neutralizes acidic chyme
 - Bile enters duodenum
 - Secreted from liver
 - Can be stored in gallbladder
 - Contains bile salts that aid in fat digestion

The Gastrointestinal Tract

- Absorption in small intestine
 - Absorption is completed within first 20% of the intestine length
 - Anatomically arranged to provide a large surface area for absorption

The Gastrointestinal Tract

- Small intestine wall
 - Villi increase surface area of epithelium
 - Contain blood vessels and lacteal for absorption of nutrients
 - Microvilli increase surface area of epithelial cells
 - Form brush border
- Crypts of Lieberkuhn
 - Epithelial cells in crypts secrete bicarbonate-rich fluid
 - Secreted in proximal small intestine
 - Absorbed in distal small intestine

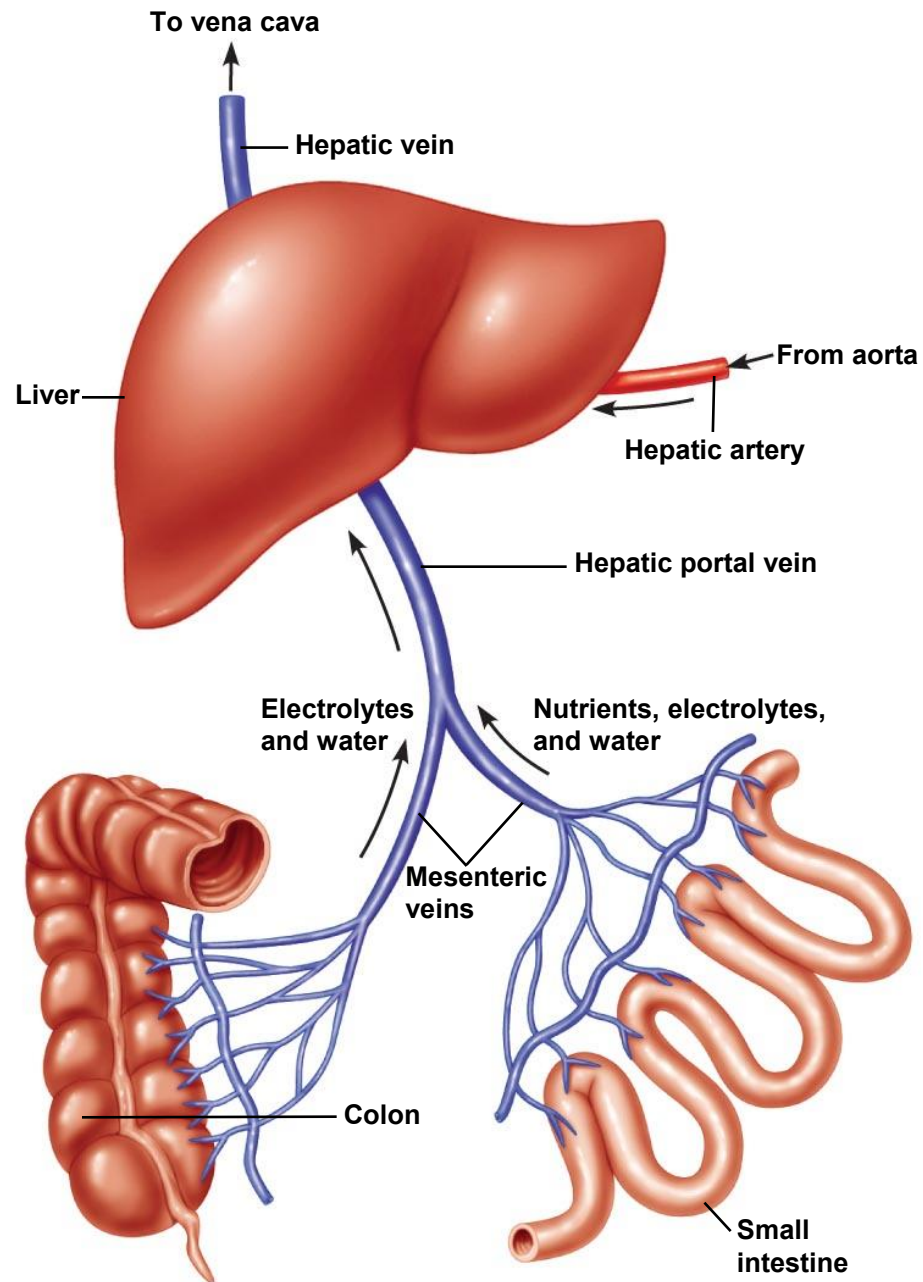
The Gastrointestinal Tract

- Role of liver in absorption
 - Absorbed nutrients travel in the blood to the liver before entering the general circulation
 - Liver functions to detoxify substances and processes certain nutrients

The Gastrointestinal Tract

- Hepatic portal system
 - Vasculature that delivers absorbed nutrients to the liver before nutrients enter the general circulation
 - Nutrients are absorbed from the small intestine into the mesenteric veins
 - Nutrients travel from the mesenteric veins to the liver via the hepatic portal vein
 - Nutrients travel from the liver to the heart (general circulation) via the hepatic vein
 - Hepatic artery provides the liver with fresh blood to supply oxygen

Figure 20.7 Circulatory route of materials absorbed in the small intestine and colon.



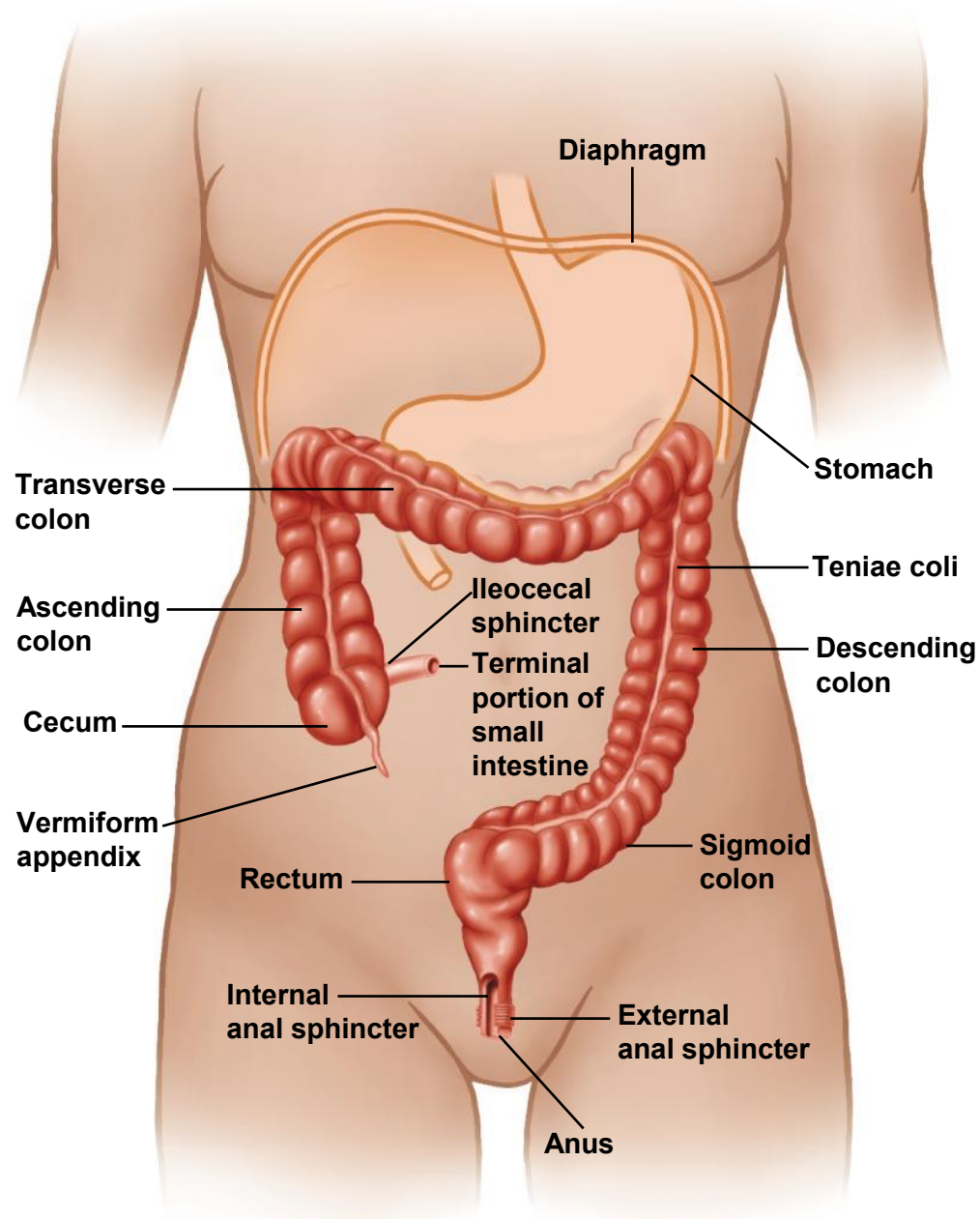
The Gastrointestinal Tract

- Large intestine
 - Large intestine consists of cecum, colon, and rectum
 - Ileocecal sphincter is between ileum and colon
 - Teniae coli are bands of longitudinal smooth muscle
- Colon
 - Hollow tube from small intestine to rectum
 - Functions
 - Concentration of wastes into feces
 - Absorption of most water
 - Storage of feces until defecation

The Gastrointestinal Tract

- Four sections of the colon
 - Ascending colon
 - Transverse colon
 - Descending colon
 - Sigmoid colon

Figure 20.8 Major anatomical features of the colon.



The Gastrointestinal Tract

- Rectum and anus
 - Colon empties into the rectum, then feces is excreted through the anus
 - Colon, rectum, and cecum make up the large intestine
 - Internal anal sphincter: smooth muscle
 - External anal sphincter: skeletal muscle
 - Relaxation of both sphincters necessary to open sphincter and excrete feces: defecation

The Accessory Glands

- Products are secreted via ducts into the lumen of the GI tract
- Four types of glands
 - Salivary glands: secrete saliva
 - Pancreas: secretes pancreatic juice
 - Acini
 - Liver: secretes bile

Figure 20.9 Generalized structure of an accessory gland.

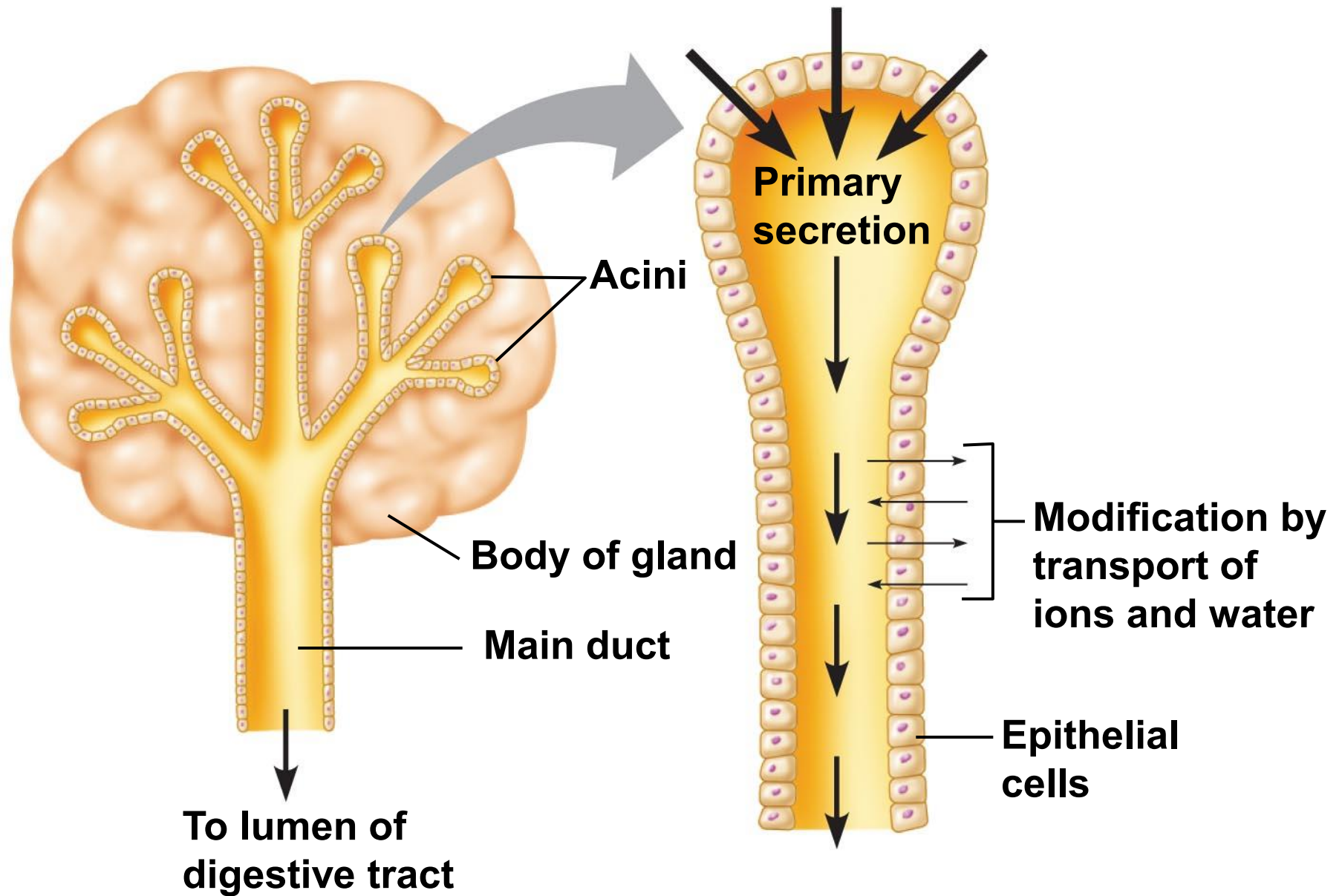
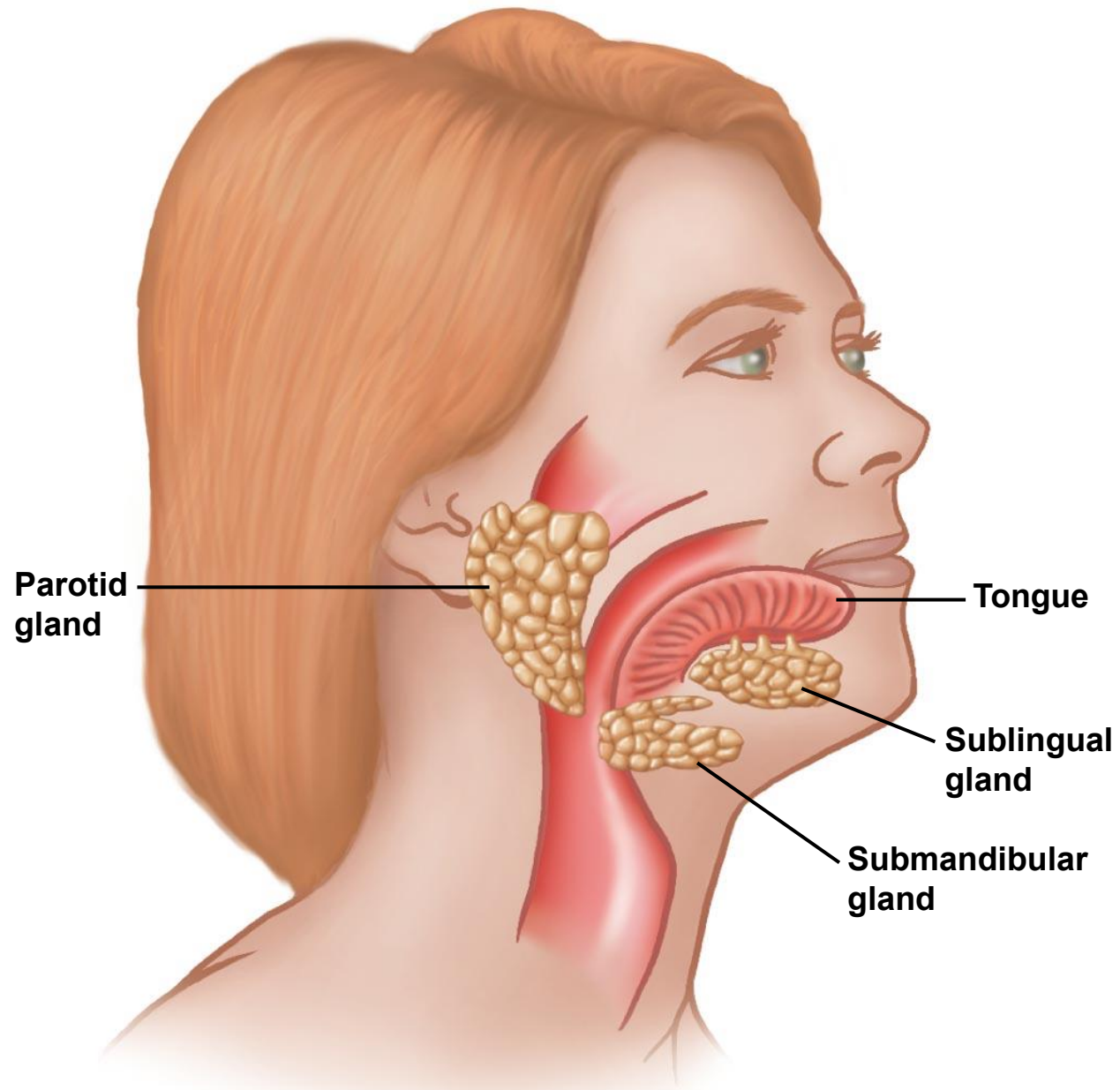


Figure 20.10 Location of the major salivary glands.



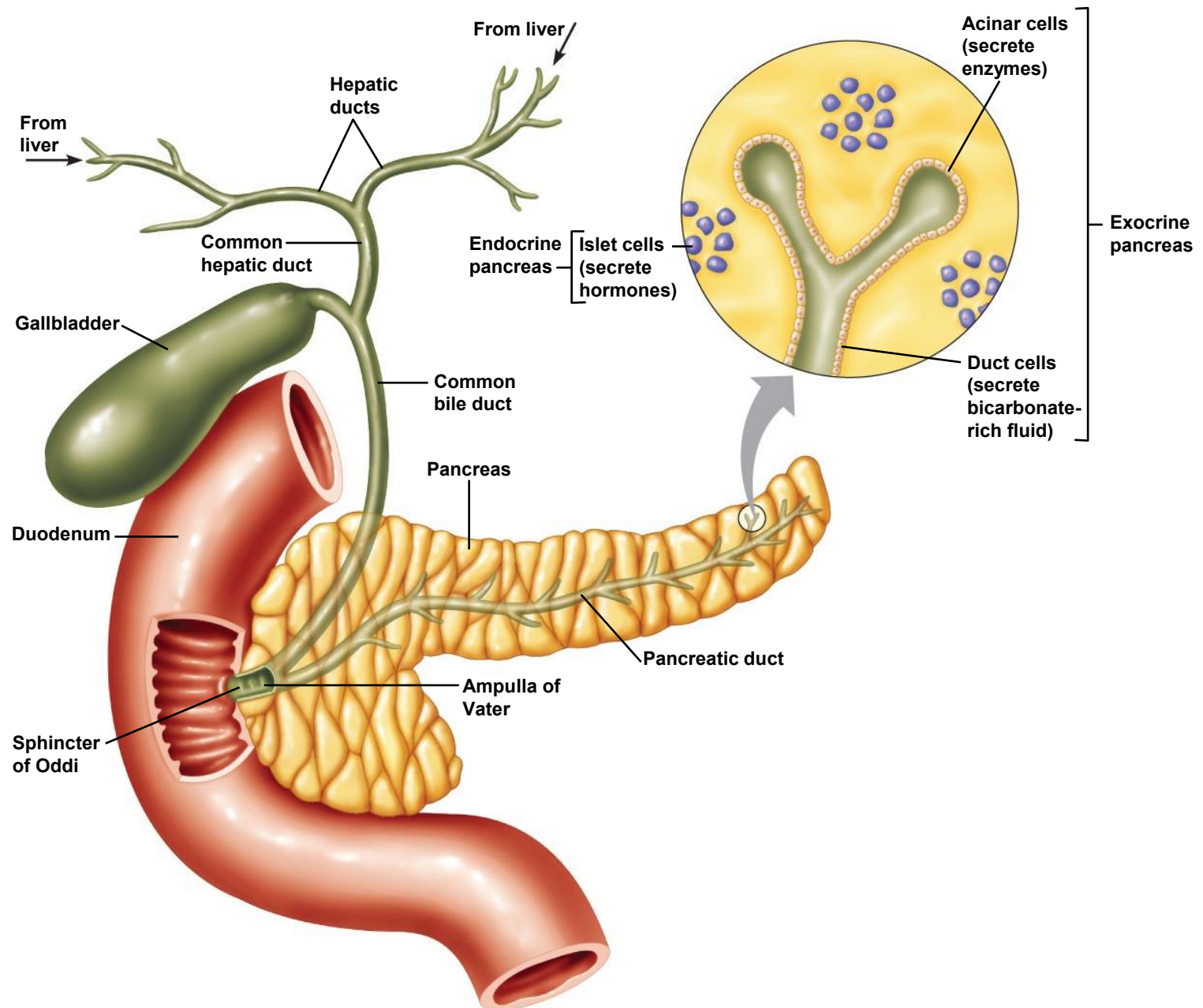
The Accessory Glands

- Characteristics of saliva
 - Rich in bicarbonate ions
 - Contains mucus
 - Enzymes present
 - Salivary amylase
 - Lysozyme

The Accessory Glands

- The pancreas
 - Has exocrine and endocrine portions
 - Exocrine portion produces pancreatic juice
 - Rich in bicarbonate
 - Relative to weight, secretes more protein than any other tissue in body
 - Enzymes present
 - Pancreatic amylase and lipases
 - Proteases
 - Nucleases

Figure 20.11 The pancreas and associated structures.



The Accessory Glands

- The liver
 - Secretes bile
 - Processes nutrients
 - Glucose → glycogen
 - Amino acids → fatty acids
 - Triglycerides and cholesterol synthesis
 - Lipoprotein synthesis

The Accessory Glands

- Liver functions
 - Removal of old red blood cells
 - Catabolizes hemoglobin
 - Generates bilirubin
 - Elimination of wastes and toxins
 - Bilirubin is eliminated as bile pigments
 - Synthesis of plasma proteins
 - Secretion and modification of hormones

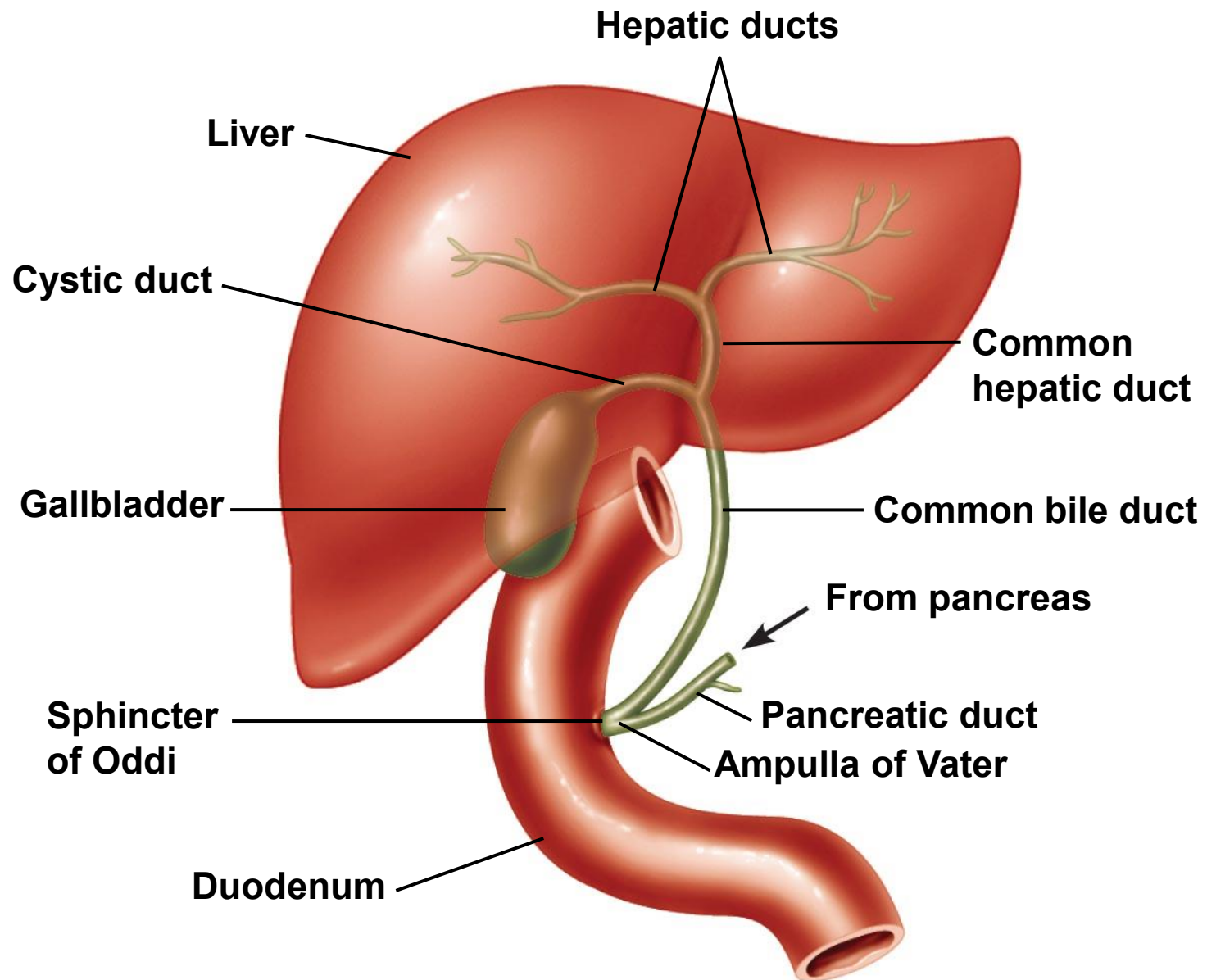
The Accessory Glands

- Biliary system: liver, gallbladder, and associated ducts
 - Liver synthesizes bile
 - Gallbladder stores bile
 - Common bile duct
 - Transports bile from liver and/or gallbladder to duodenum
 - Joins with pancreatic duct to form ampulla of Vater
 - Sphincter of Oddi: regulates flow from pancreas and gallbladder to duodenum

The Accessory Glands

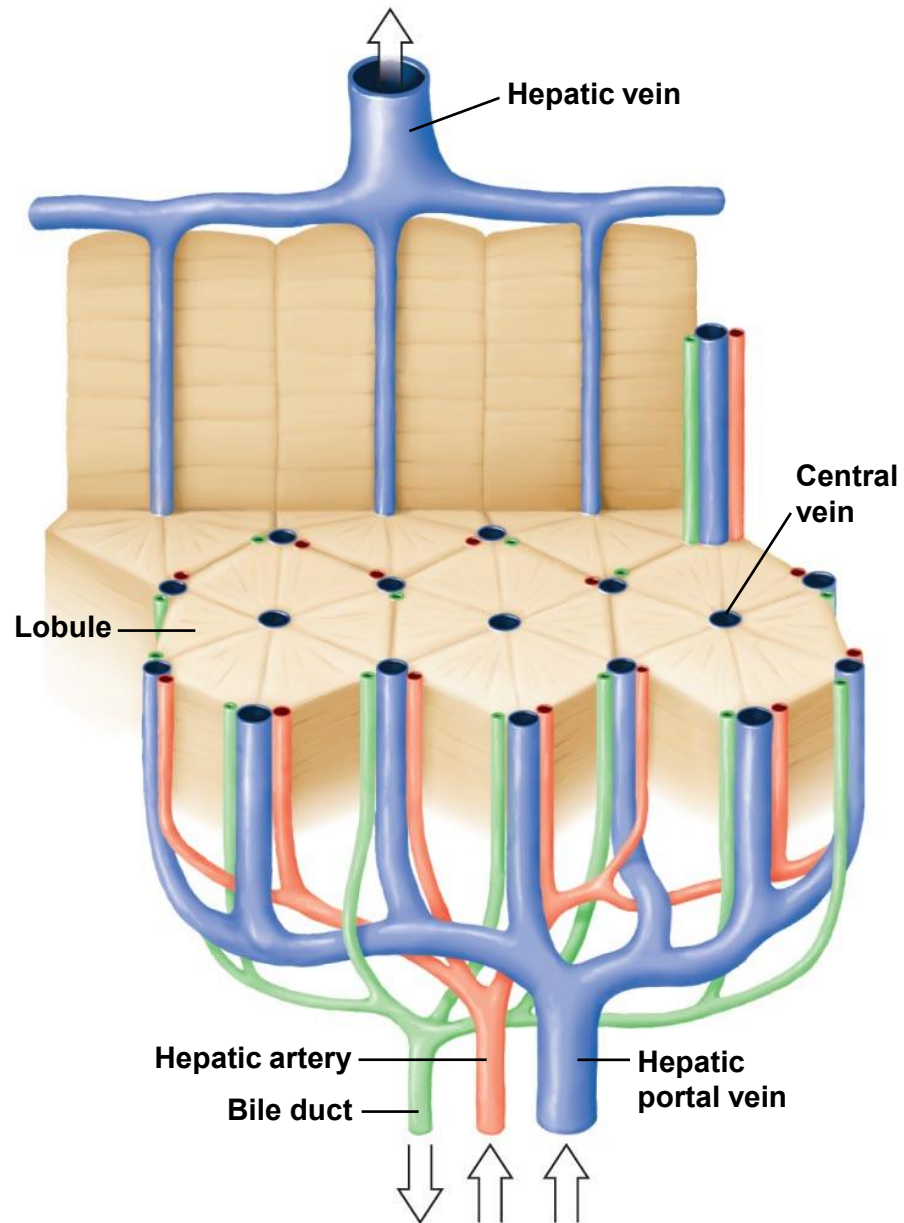
- Liver anatomy
 - Materials to make bile are taken up from blood in the sinusoids into hepatocytes
 - Hepatocytes secrete bile into bile canaliculi, on the side opposite of the sinusoids
 - Canaliculi drain into bile ducts
 - Bile ducts drain into common hepatic duct

Figure 20.12a Structures of the biliary system.



(a)

Figure 20.12b Structures of the biliary system.



(b)

TABLE 20.1 Gastrointestinal System Organs and Their Functions

Organ	Functions
Mouth	Mechanical breakdown of food; mixing of food with saliva; initiation of chemical digestion of carbohydrates by salivary amylase
Pharynx	Conduction of food to esophagus
Esophagus	Conduction of food to stomach
Stomach	Mechanical breakdown of food; secretion of acid, pepsinogen, and intrinsic factor; initiation of chemical digestion of proteins by pepsin; secretion of gastrin into bloodstream; transformation of food into chyme
Small intestine	Chemical digestion of all nutrient classes by pancreatic enzymes and brush border enzymes; absorption of digestive end-products, water, ions, and vitamins; secretion of enterogastrones into bloodstream; secretion of bicarbonate-rich fluid
Colon	Absorption of ions and water; transformation of chyme into feces; storage of feces
Rectum	Storage of feces prior to elimination
Anus	Control of defecation
Salivary glands	Secretion of saliva (contains amylase, mucus, bicarbonate, and lysozyme)
Pancreas	Secretion of pancreatic juice (contains digestive enzymes and bicarbonate)
Liver	Secretion of bile (contains bile salts and bicarbonate); processing of absorbed nutrients
Gallbladder	Storage and concentration of bile

20.3 Digestion and Absorption of Nutrients and Water

- Carbohydrates
- Proteins
- Lipids
- Absorption of vitamins
- Absorption of minerals
- Absorption of water

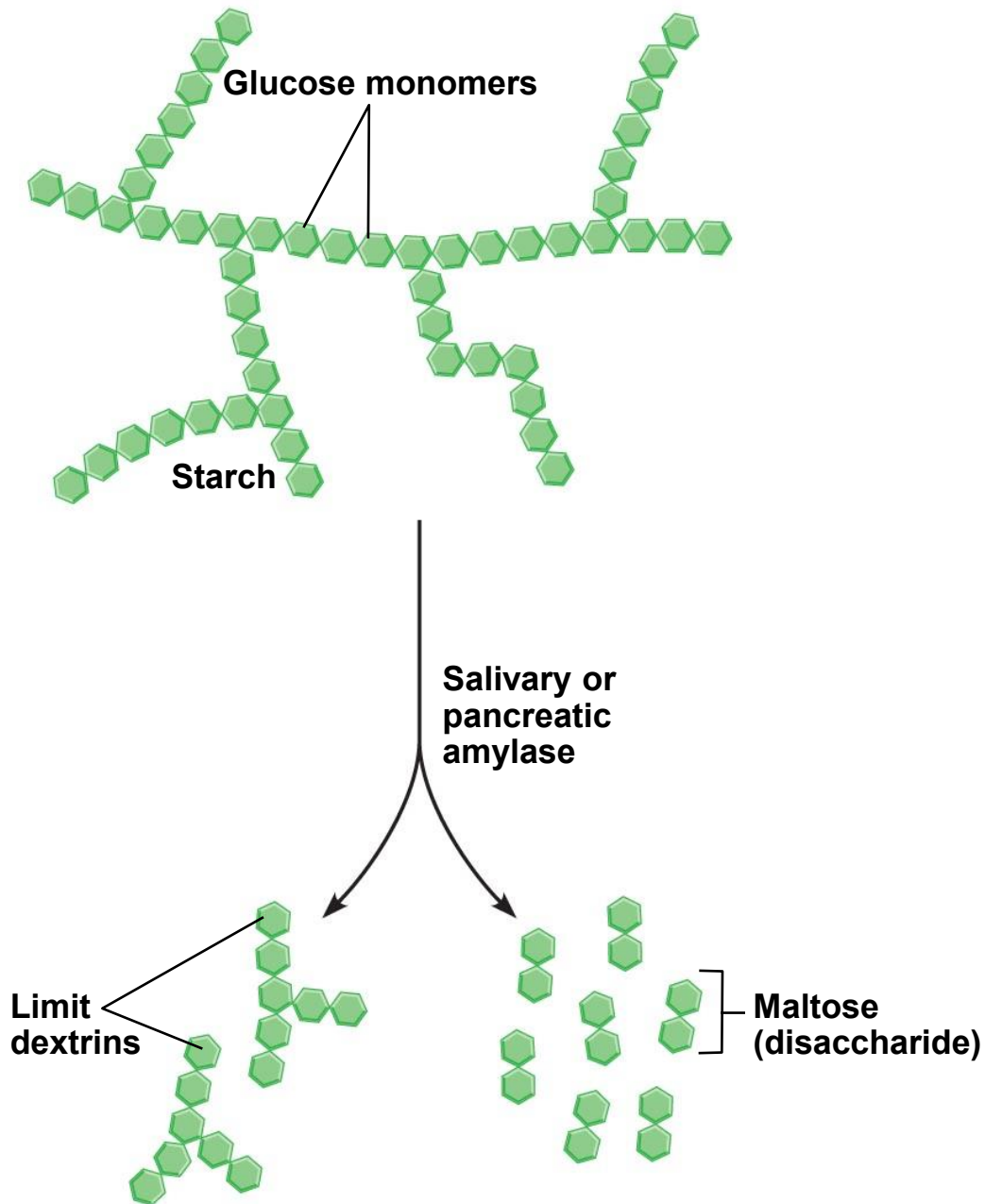
Carbohydrates

- Typical diet: 500 grams/day carbohydrates
- Most consumed as disaccharides or polysaccharides
 - Sucrose
 - Lactose
 - Maltose
 - Starch
 - Glycogen
 - Cellulose (fiber; cannot be digested)

Carbohydrates

- Carbohydrate digestion
 - Only monosaccharides are absorbed
 - Disaccharides and polysaccharides must be digested to monosaccharides
 - Digestion of starch
 - Enzyme of digestion
 - Salivary amylase
 - Pancreatic amylase
 - End-product
 - Disaccharides (maltose)
 - Limit dextrins

Figure 20.13 Digestion of starch by salivary or pancreatic amylase.



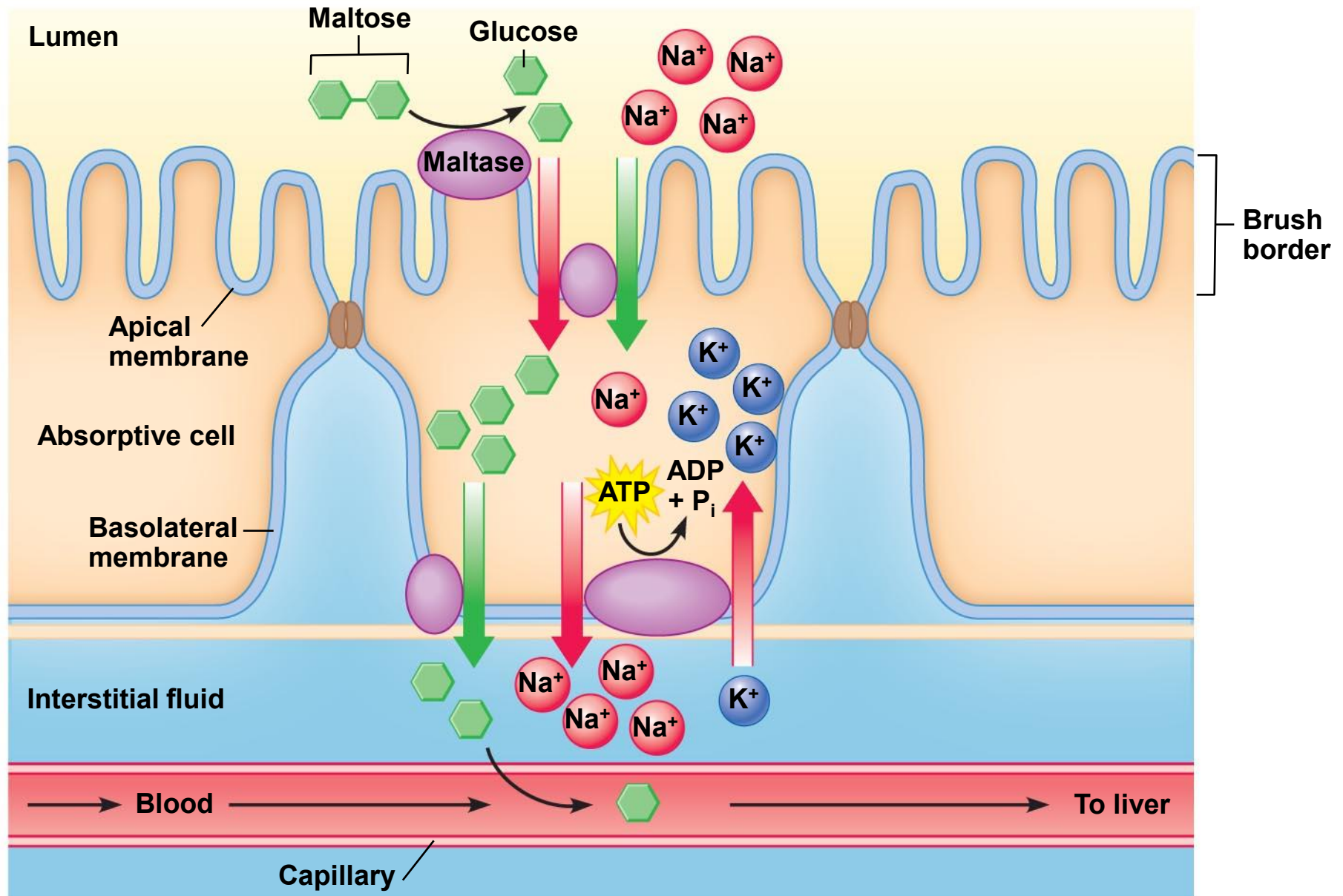
Carbohydrates

- Enzymes of digestion
 - Dextrinase: limit dextrins $\rightarrow$ glucose
 - Glucoamylase: polysaccharides $\rightarrow$ glucose
 - Sucrase: sucrose $\rightarrow$ fructose + glucose
 - Lactase: lactose $\rightarrow$ galactose + glucose
 - Maltase: maltose $\rightarrow$ 2 glucose
- Location of enzymes: brush border of small intestine
 - Brush border enzymes

Absorption of Carbohydrates

- Absorption: transport of nutrients from lumen to blood
- Glucose and galactose are absorbed in two ways:
 - Secondary active transport across apical membrane
 - Facilitated diffusion across basolateral membrane
- Fructose is absorbed in one way:
 - Facilitated diffusion across both membranes

Figure 20.14 Digestion and absorption of maltose.



Proteins

- Typical diet: 125 grams/day protein
- Digestion and absorption of proteins
 - Consumed in diet
 - Secreted into lumen of intestinal tract
 - Sloughed off with cells lining intestinal tract
- Protein digestion products
 - Amino acids
 - Dipeptides
 - Tripeptides

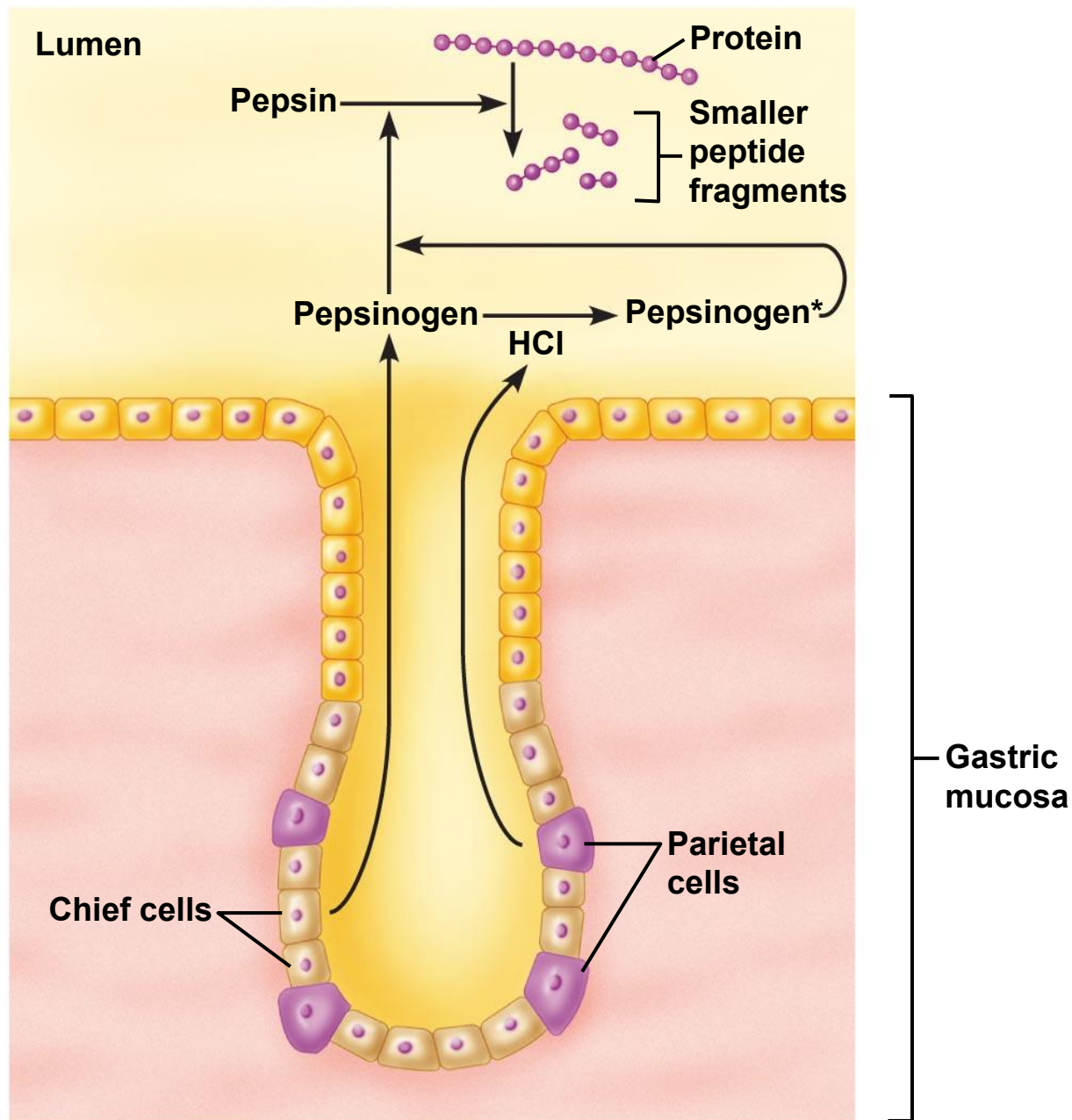
Proteins

- Proteases
 - Endopeptidases
 - Split polypeptides at interior peptide bonds into small peptide fragments
 - Exopeptidases
 - Cleave amino acids from one end of polypeptide
 - Zymogens
 - Inactive storage form of proteases
 - Stored in zymogen granules
 - Secreted by exocytosis
 - Activated by proteolysis

Proteins

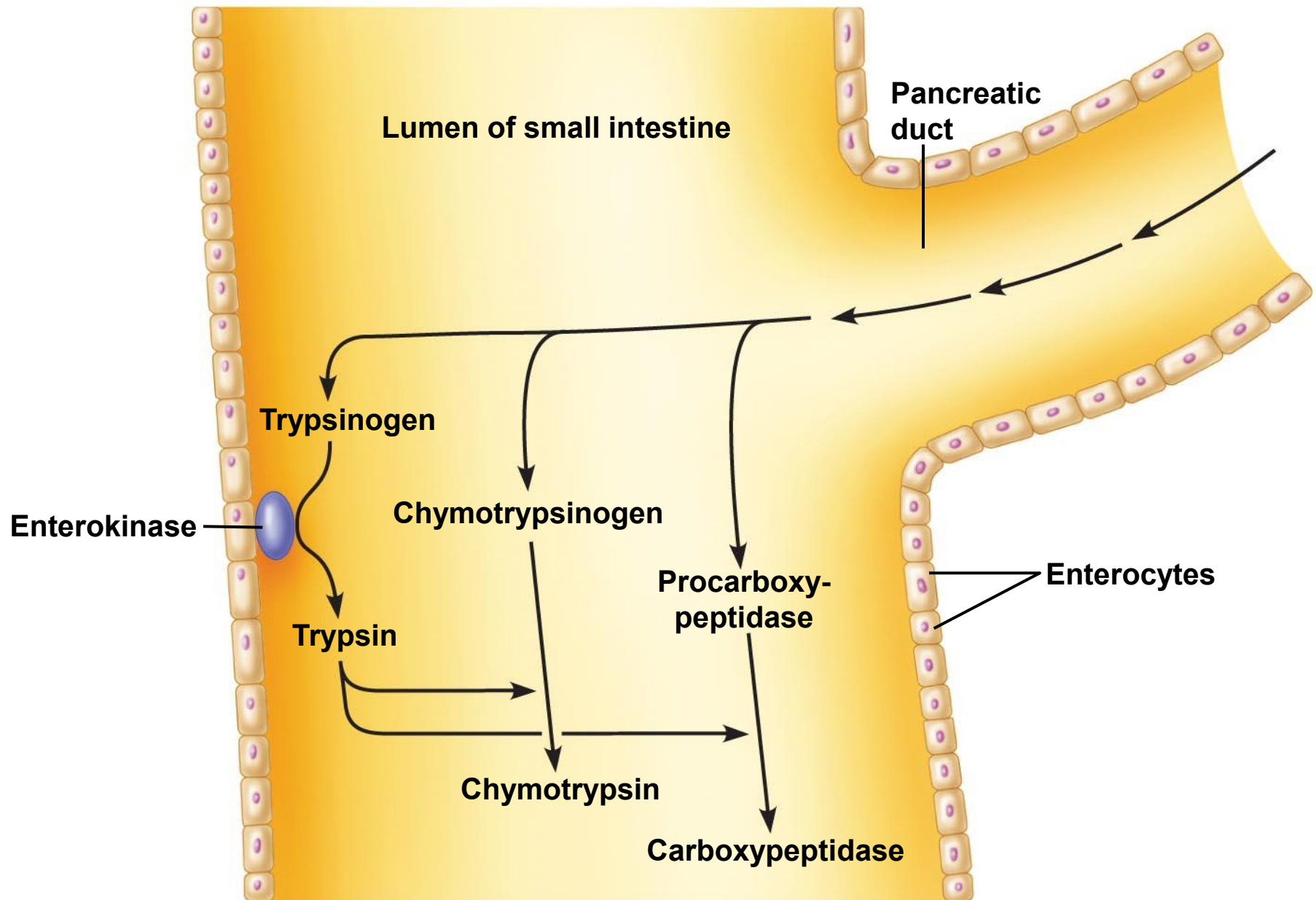
- Protein digestion begins in stomach
- Enzyme: pepsin
- Inactive (secreted) form: pepsinogen
- Activated by acid
- Pepsin: activation and activity
 - Chief cells secrete pepsinogen
 - Pepsinogen: inactive form
 - HCl cleaves pepsinogen to pepsin
 - Pepsin: active form
 - Parietal cells secrete HCl

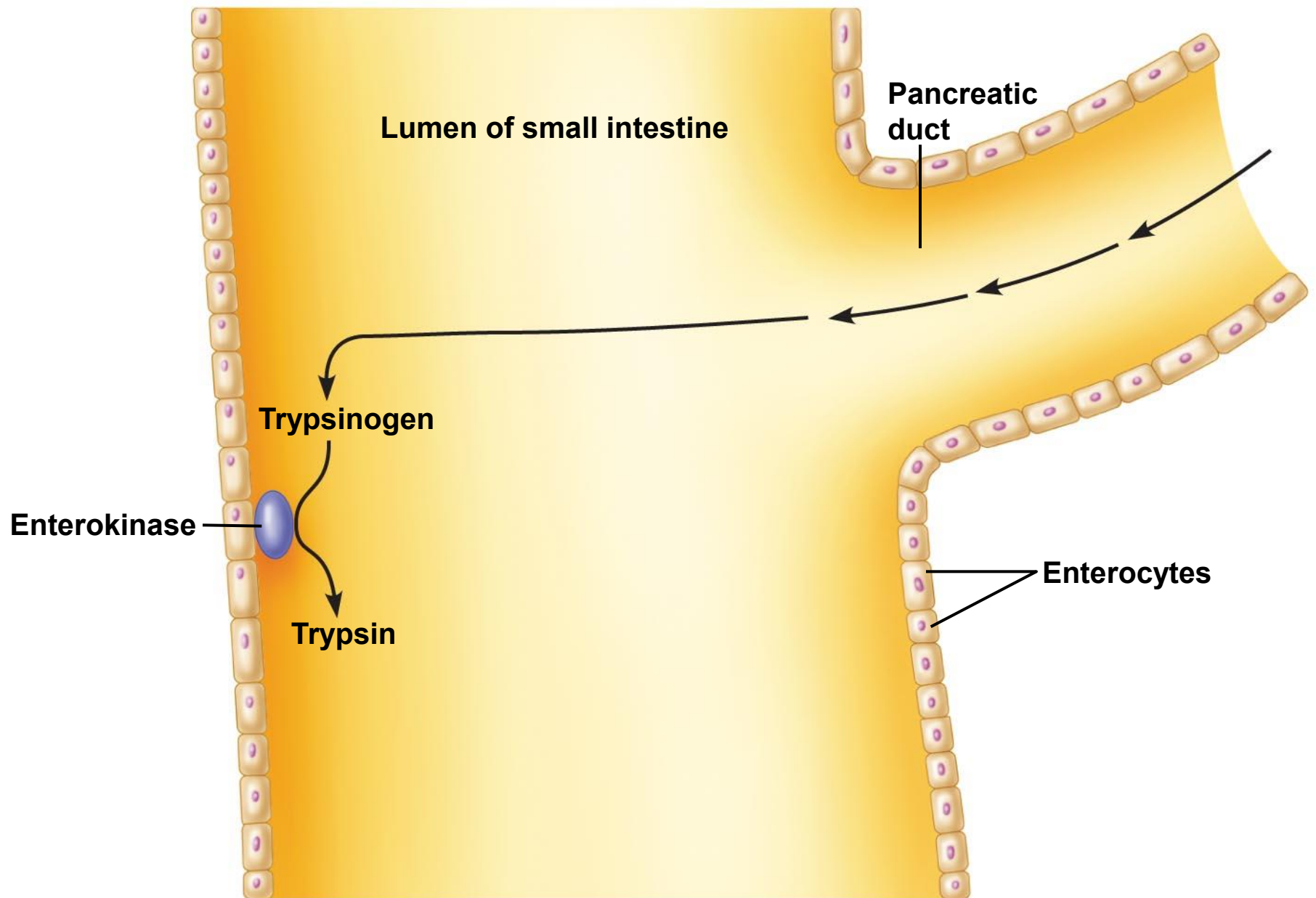
Figure 20.15 Activation and activity of pepsin in the stomach.

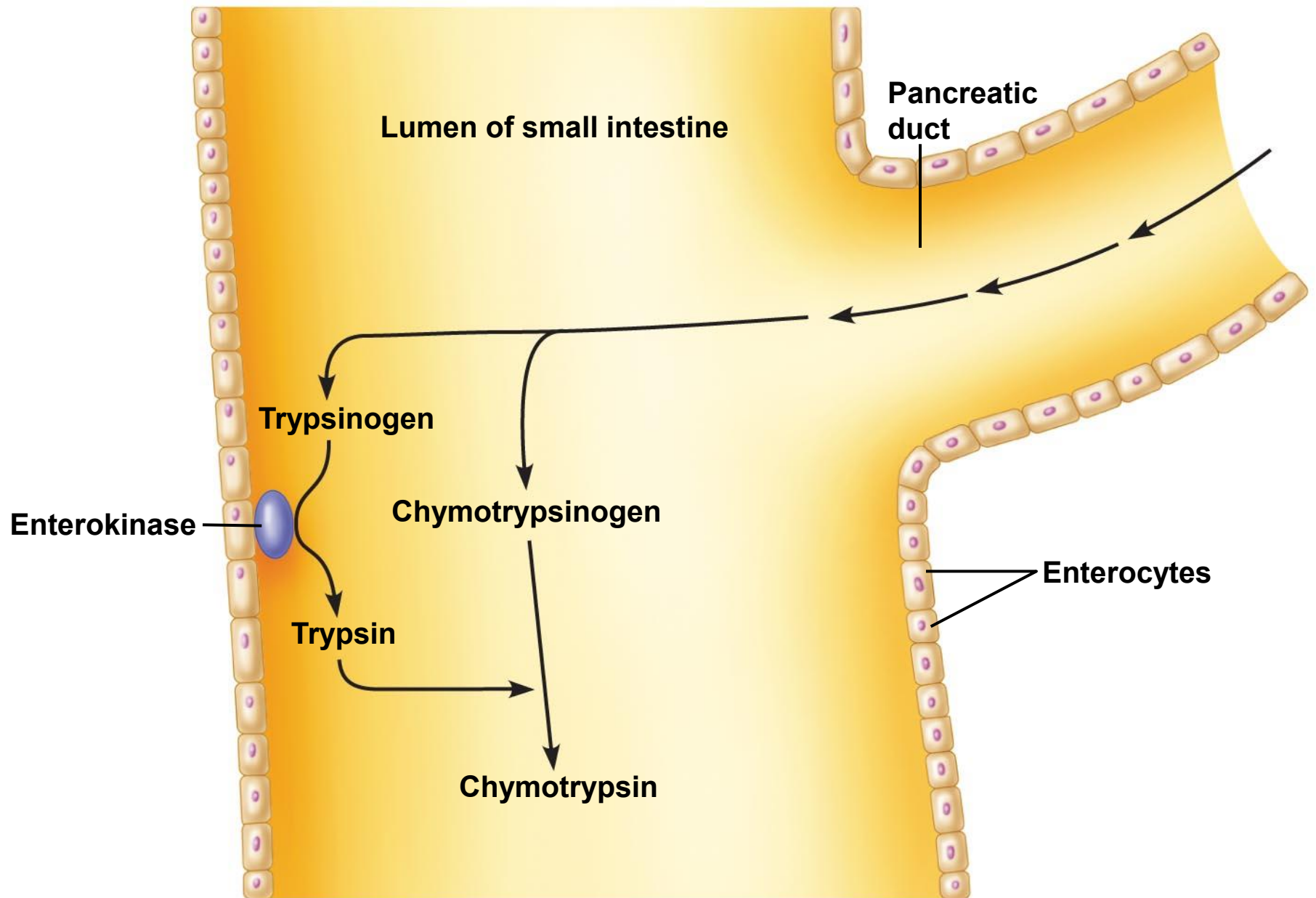


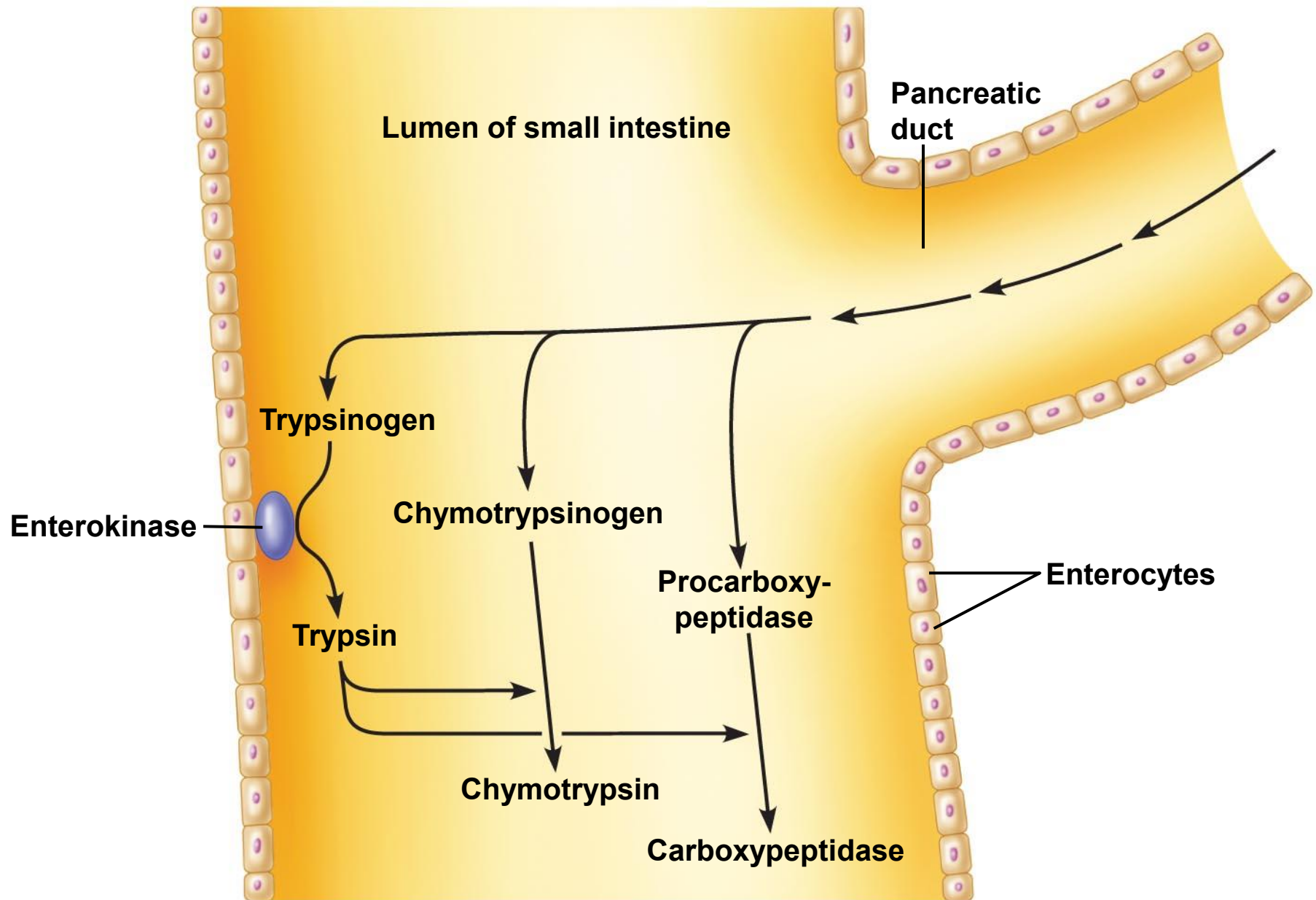
Proteins

- Protein digestion in small intestine
 - Pancreatic proteases
 - Trypsin
 - Chymotrypsin
 - Carboxypeptidase
 - Brush border proteases
 - Aminopeptidase
 - Enterokinase









Proteins

- Absorption of amino acids
 - Cross the apical membrane by sodium-linked secondary active transport or facilitated diffusion
 - Cross the basolateral membrane by facilitated diffusion
- Absorption of dipeptides and tripeptides
 - Cross the apical membrane by active transport
 - Broken down inside cells to amino acids
 - Amino acids cross the basolateral membrane by facilitated diffusion

Lipids

- Typical diet: 50 grams/day lipids
 - 90% triglycerides
- Lipids face special problem in digestion and absorption
 - Not water soluble
 - Do not mix with stomach or intestinal contents
 - Form fat droplets

Lipids

- Lipid digestion
 - Enzymes of digestion: lipases
 - Secreted from pancreas
 - Lipases can only act on molecules near the edge of a fat droplet
 - Bile salts increase the surface area of droplets by breaking each large droplet into several small droplets: emulsification

Lipids

- Bile salts
 - Synthesized in liver from cholesterol
 - Secreted in bile to duodenum
 - Amphipathic molecule
 - Emulsify fat

Figure 20.17 A representative bile salt.

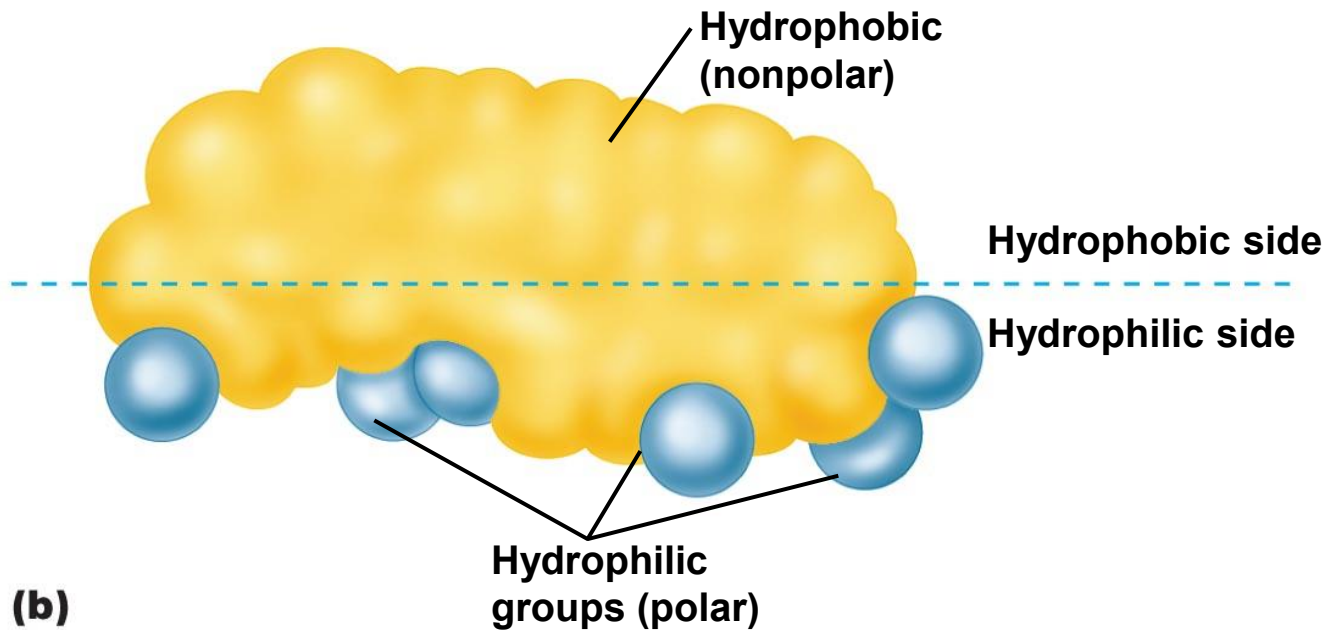
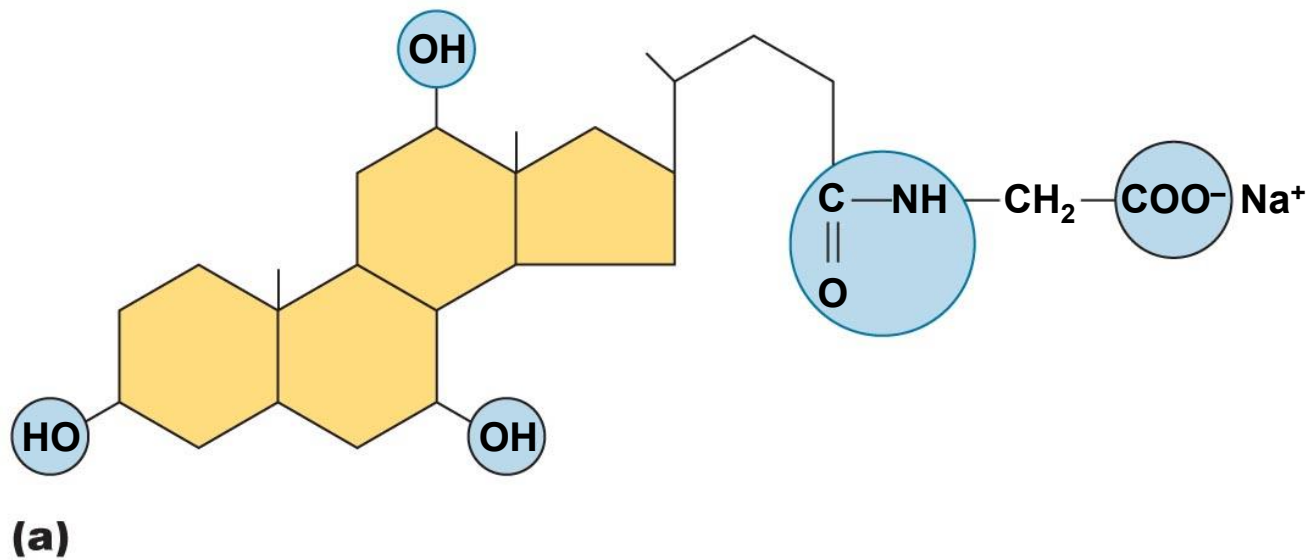
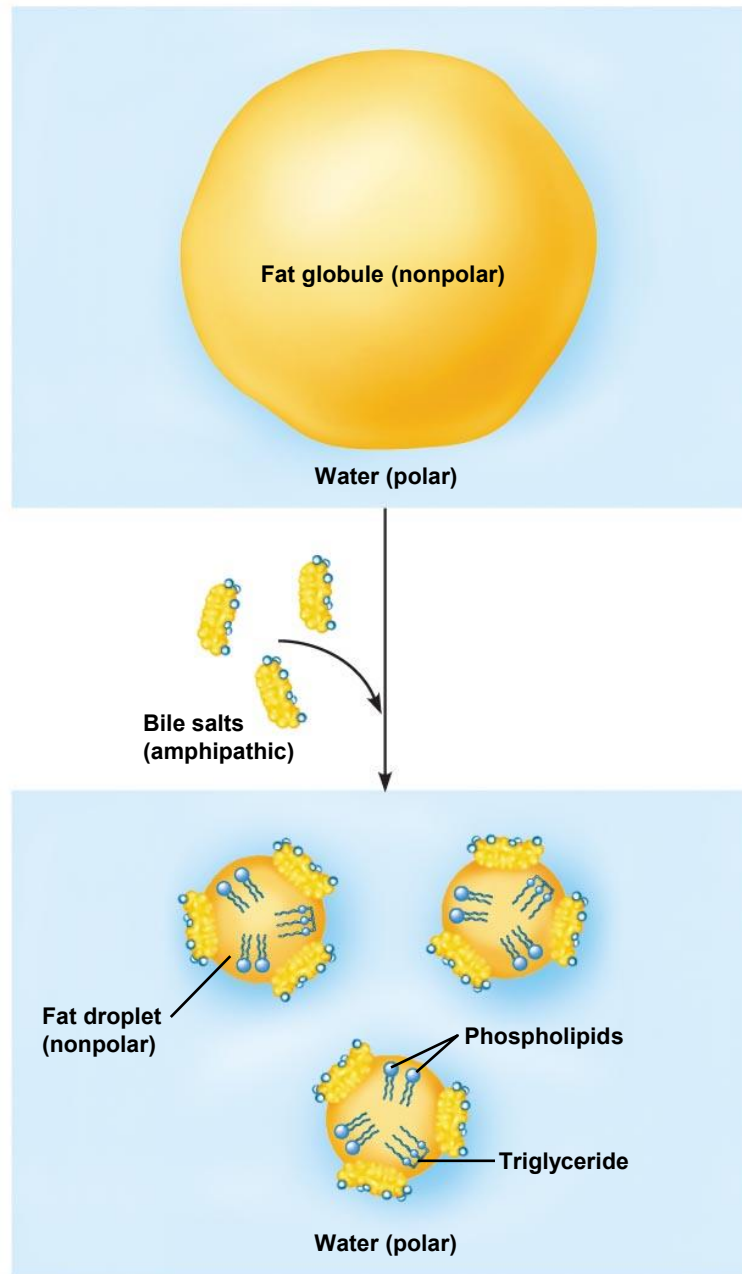


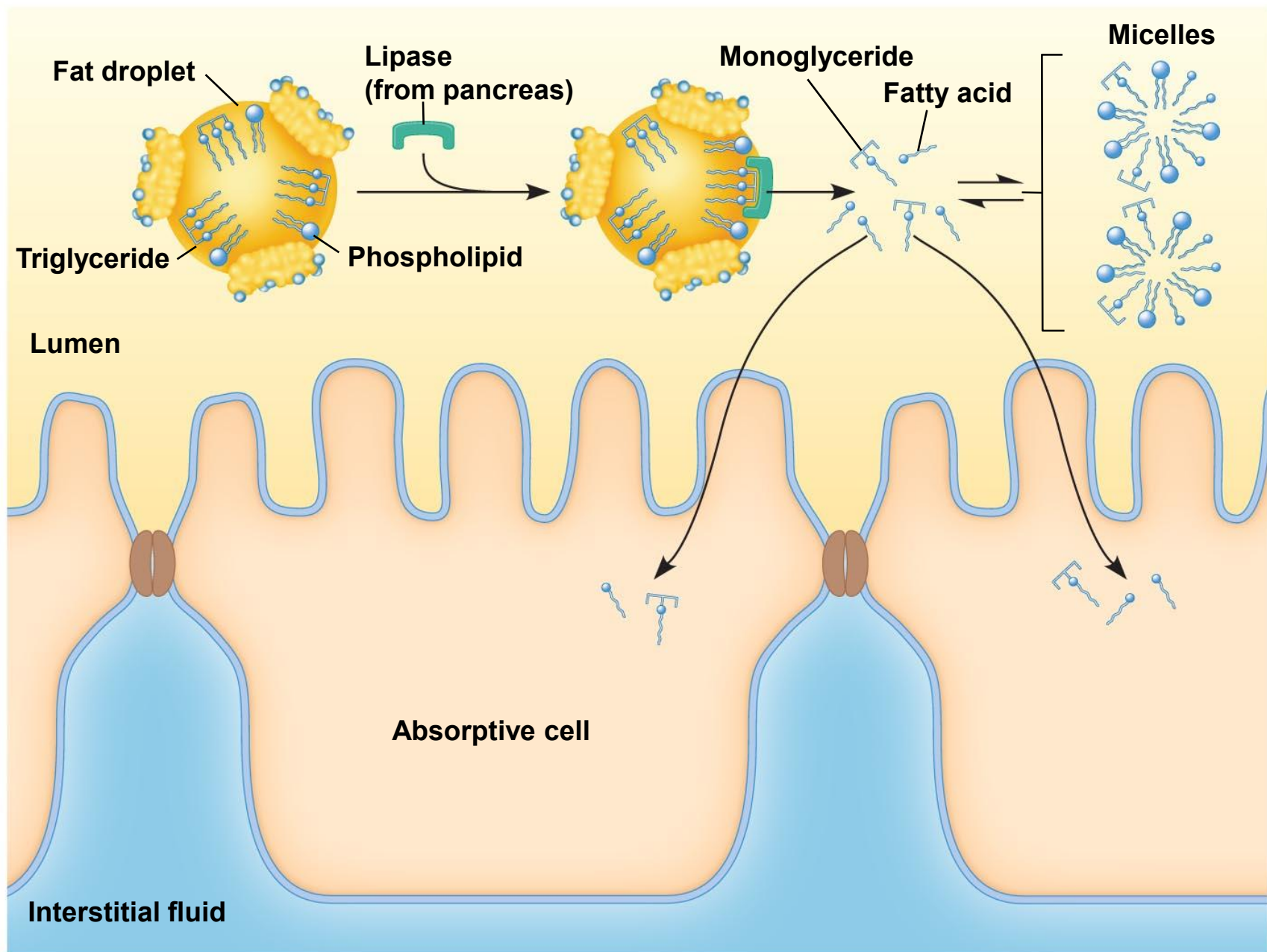
Figure 20.18 Emulsification of a fat globule by bile salts.



Lipids

- Digestion by pancreatic lipases
 - Triglycerides $\rightarrow$ monoglyceride + 3 fatty acids
 - Some fatty acids and monoglycerides are absorbed
 - Others form micelles
 - Equilibrium between micelles and the fatty acids and monoglycerides formed by action of lipases

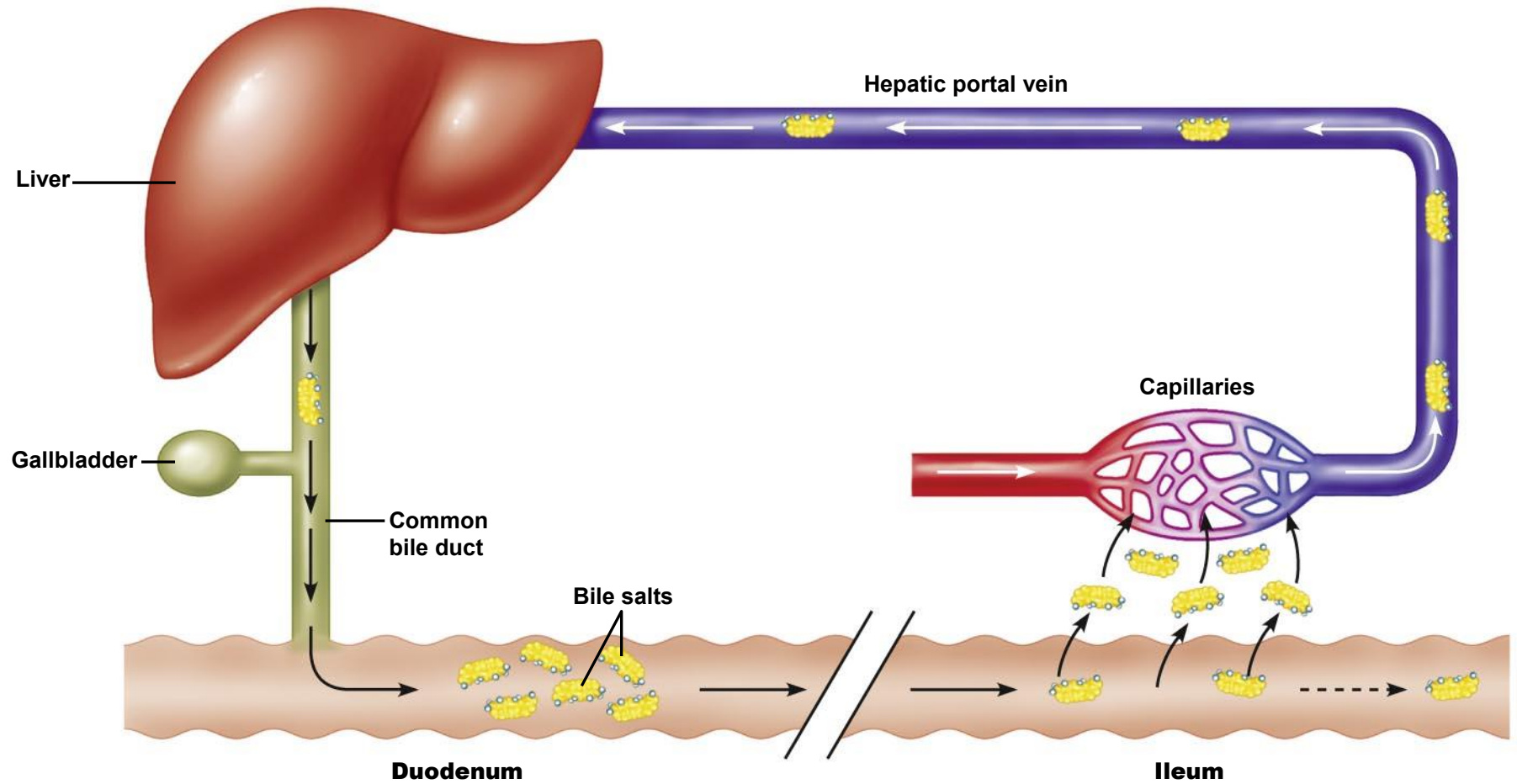
Figure 20.19 Liberation of fatty acids and monoglycerides from fat droplets by lipases.

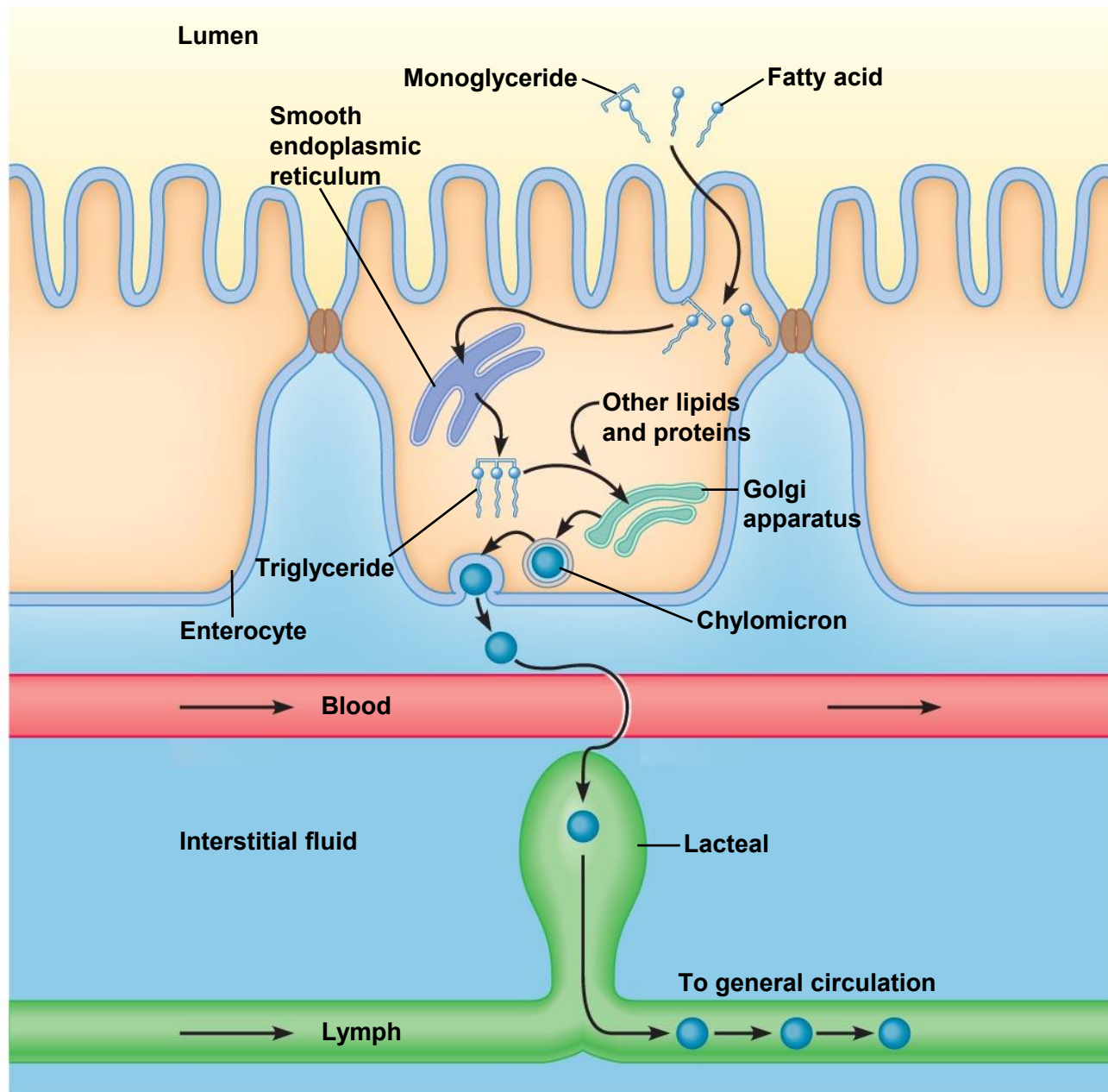


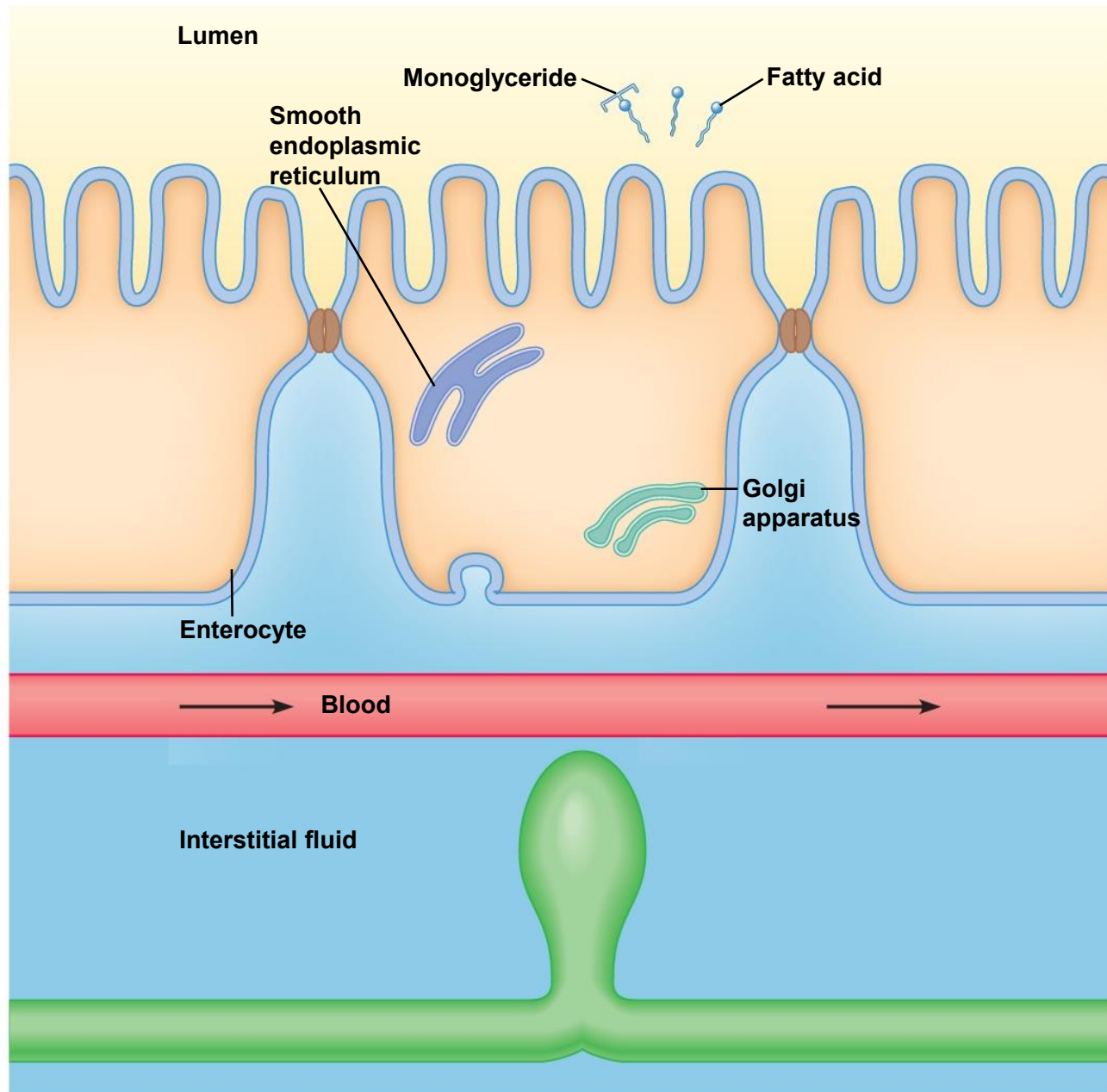
Lipids

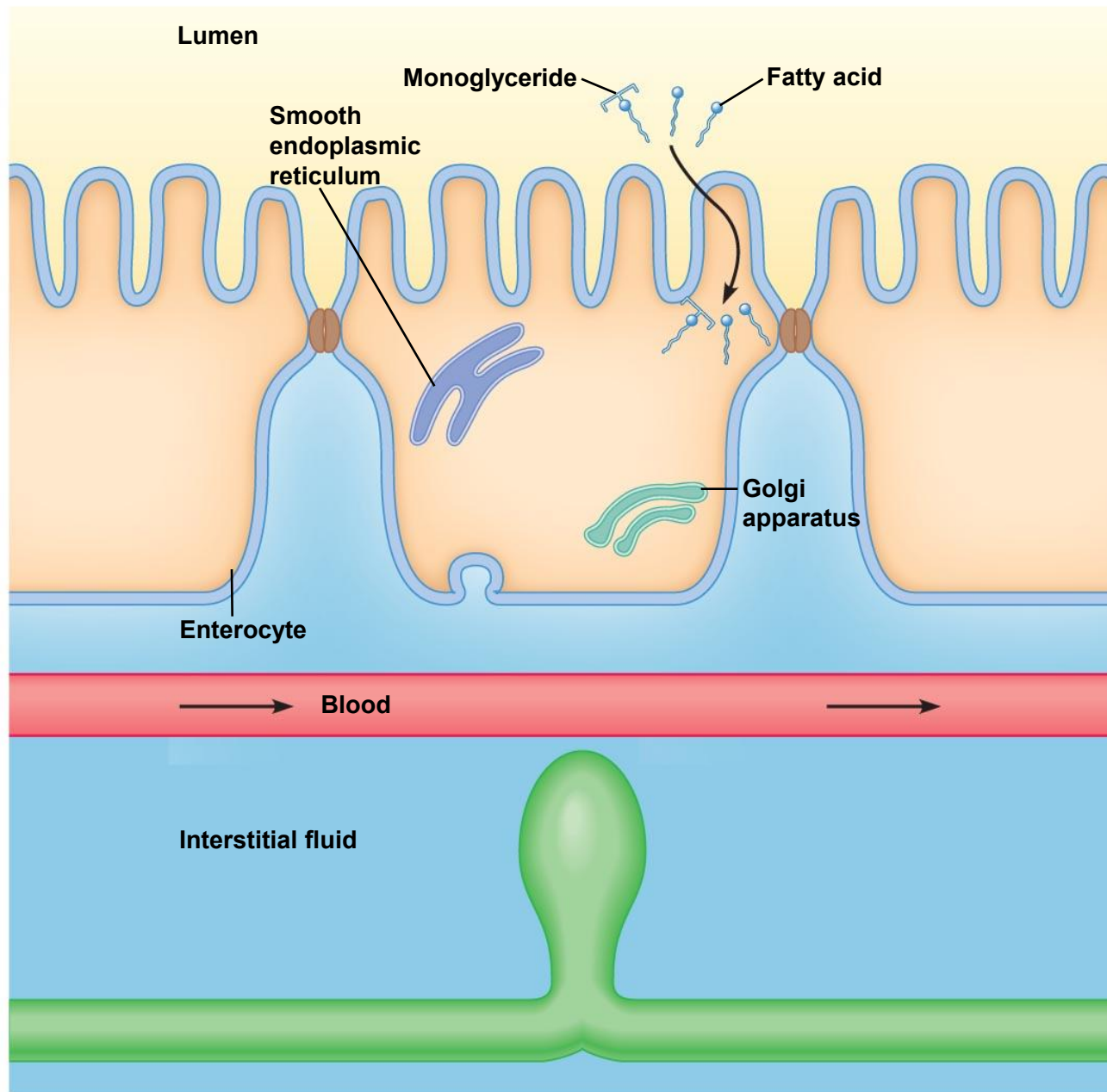
- Absorption of monoglycerides and fatty acids
 - Equilibrium between free fatty acids and monoglycerides and those inside micelles
 - Free form can be absorbed by simple diffusion across epithelium
 - Inside epithelial cell
 - Enter smooth ER and re-form triglycerides and other lipids
 - Lipids enter Golgi apparatus to be packaged into chylomicrons
 - Chylomicrons are secreted by exocytosis into interstitial fluid
 - Chylomicrons enter lymphatic system via lacteal

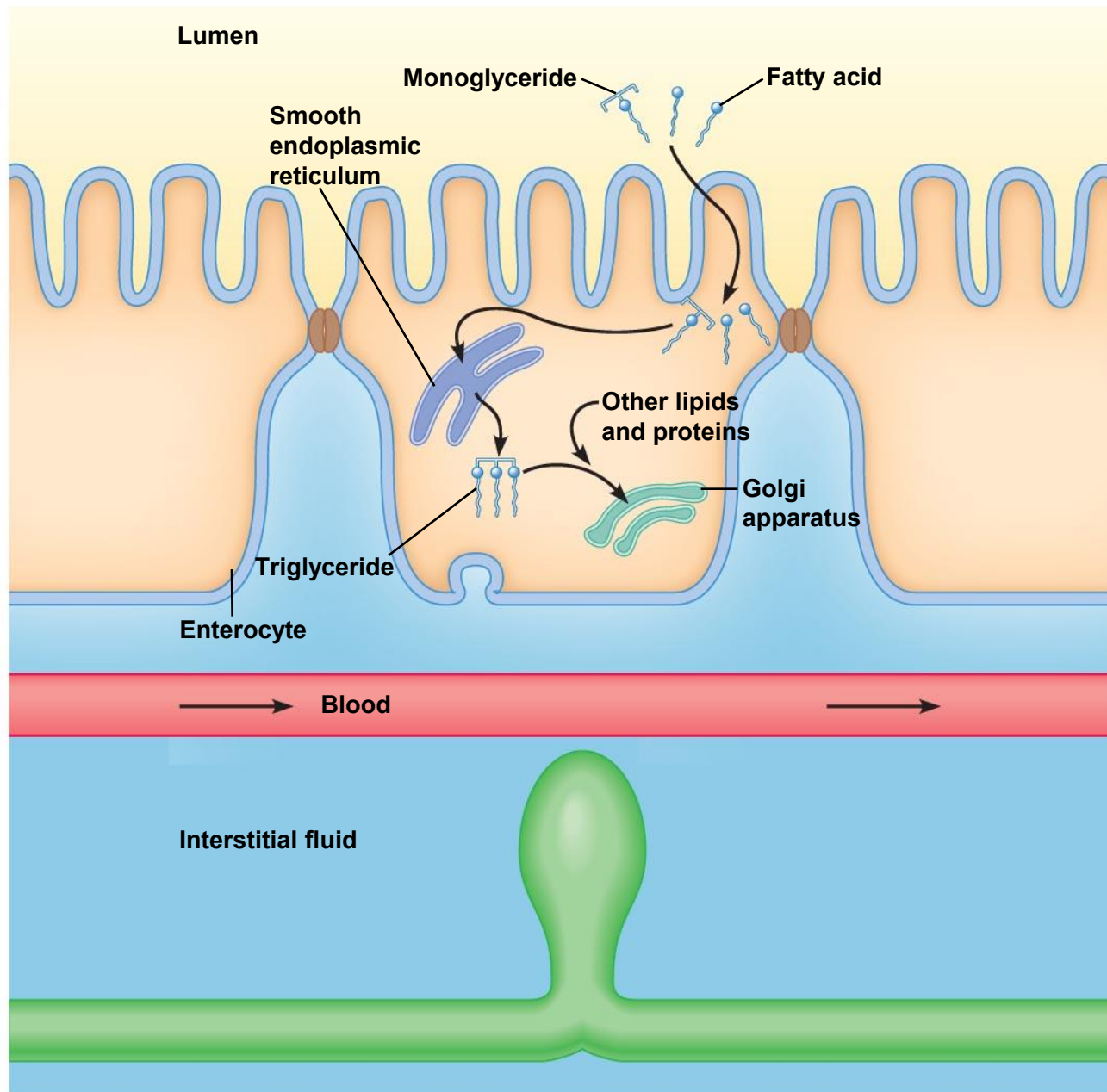
Figure 20.20 The enterohepatic circulation.

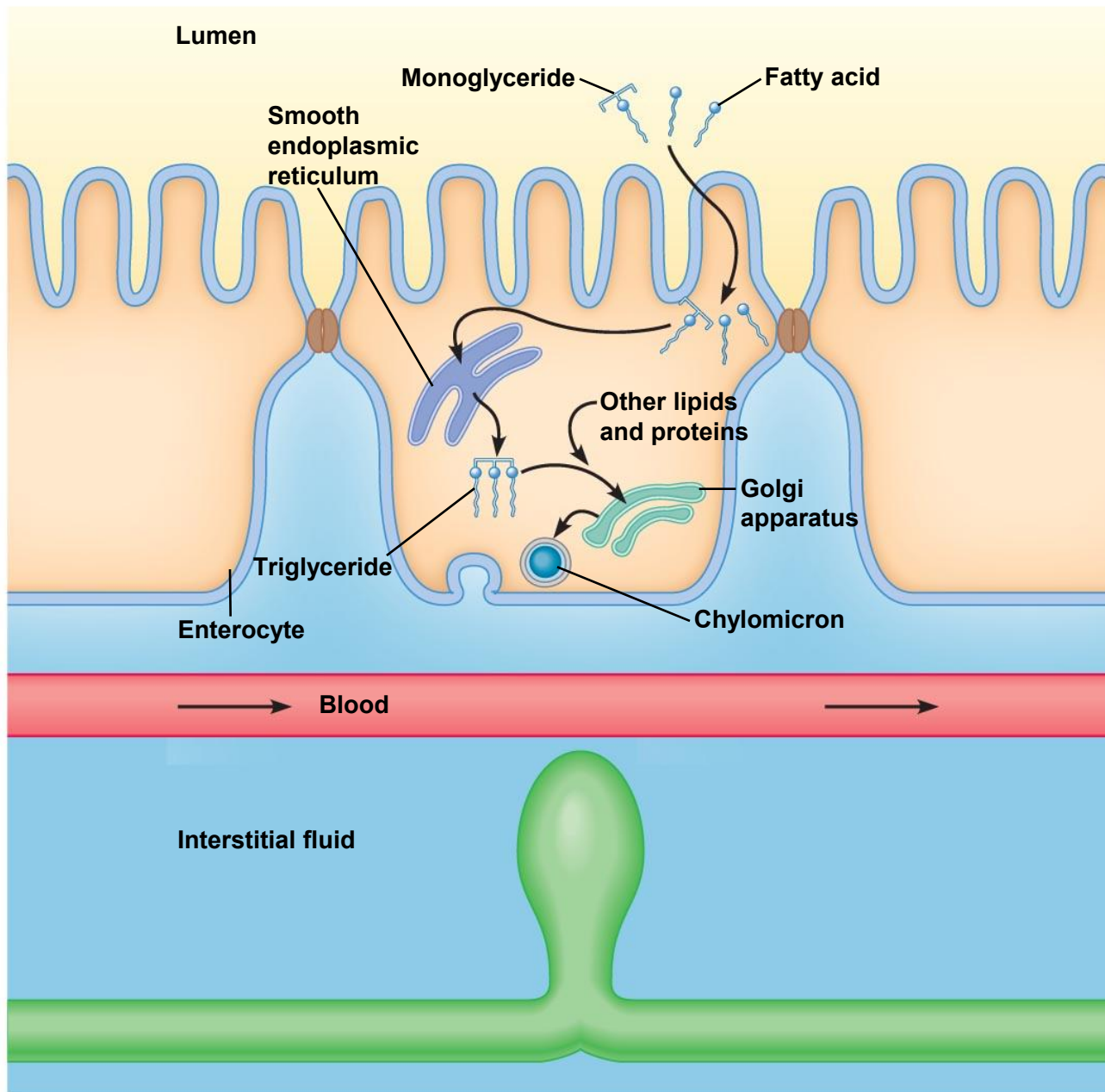


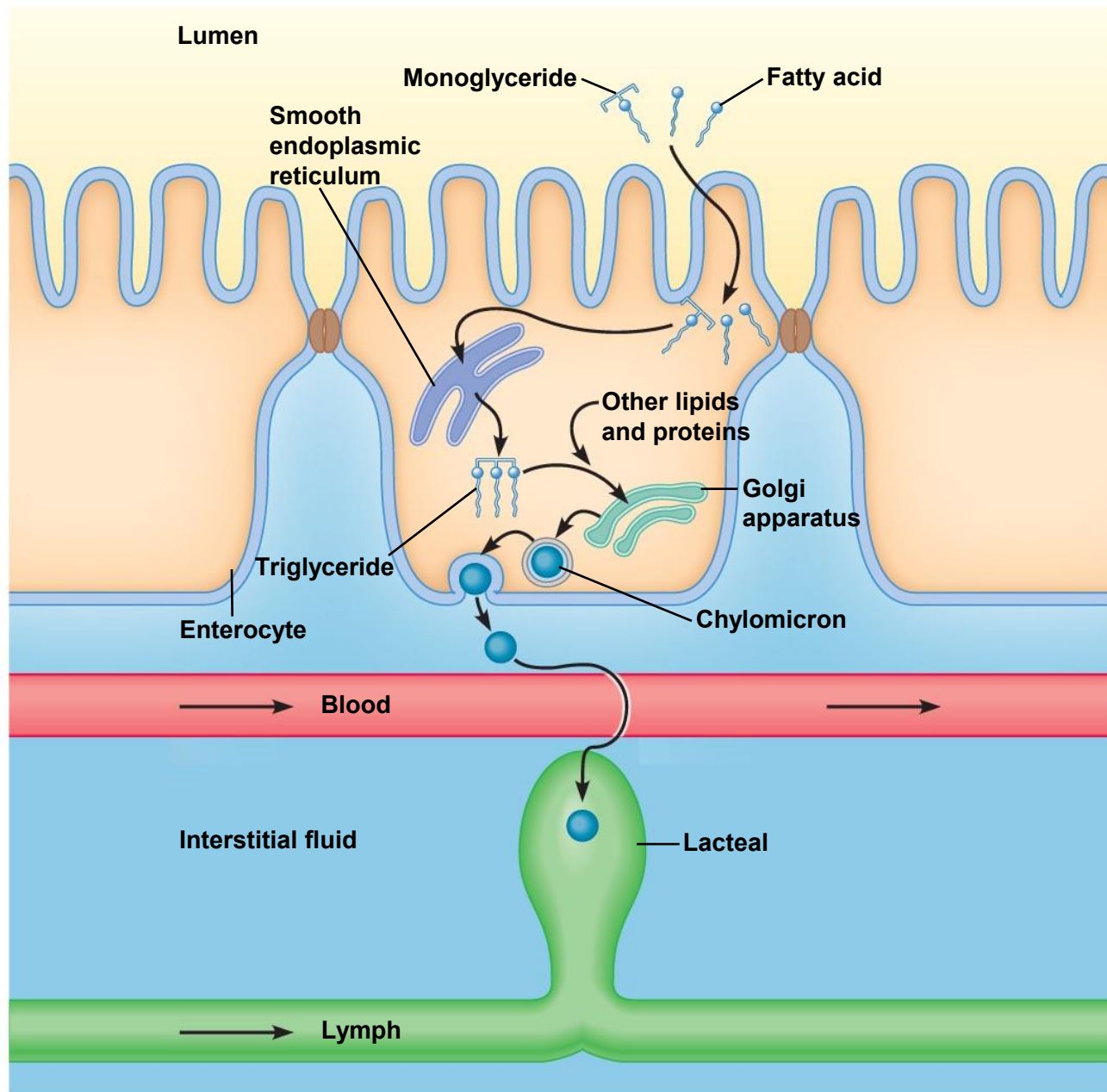


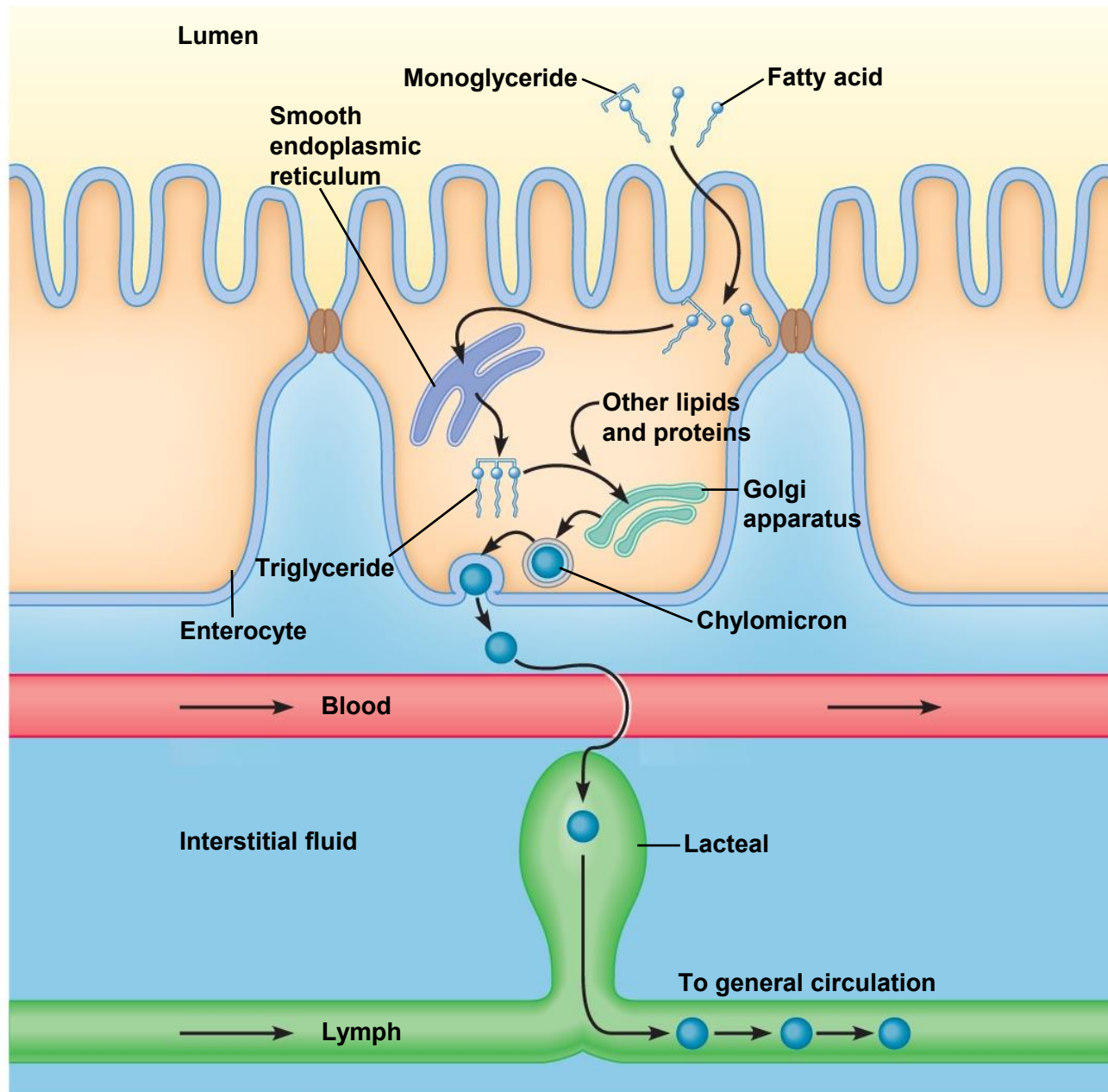












Absorption of Vitamins

- Fat-soluble vitamins (A, D, E, and K)
 - Absorbed with lipids
 - Dissolve in lipid droplets, micelles, and chylomicrons
- Water-soluble vitamins
 - Require special transport proteins
- Vitamin B₁₂
 - Absorbed only if bound to intrinsic factor

Absorption of Minerals

- Sodium
 - Solvent drag with water reabsorption
 - Actively absorbed in jejunum, ileum, and colon
- Chloride
 - Passively follows sodium absorption
- Potassium
 - Passively absorbed
 - Secreted when lumenal concentrations are very low

Absorption of Minerals

- Absorption and secretion of bicarbonate
 - Jejunum
 - Bicarbonate ions are passively absorbed
 - Ileum and colon
 - Bicarbonate are secreted in exchange for chloride ions

Absorption of Minerals

- Calcium
 - Actively absorbed in duodenum and jejunum
 - Binds to brush border protein: calcium-binding protein
 - Transported into epithelial cells
 - Transported out of cells across the basolateral membrane by the Ca^{2+} pump
 - $1,25\text{-(OH)}_2\text{D}_3$ increases calcium absorption by increasing the concentration of calcium-binding protein

Absorption of Minerals

- Iron

- Transferrin is secreted by enterocytes into the lumen of the small intestine
- Transferrin binds iron
 - Transferrin–iron complex binds receptor
 - Taken into cells by receptor-mediated endocytosis
- Some iron is stored in enterocytes as ferritin
- Some iron is transported into blood: bound to transferrin

Absorption of Water

- Water in GI tract
 - Secretions: 7 liters/day
 - Intake: 2 liters/day
- Water absorption is passive
- Water follows absorption of solutes by osmosis